

Introduction to Tensor Networks

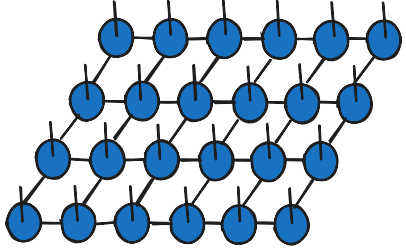


Ema Puljak

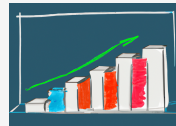
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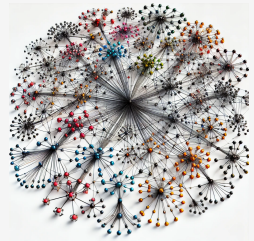
TENSOR NETWORKS



$|\Psi\rangle$ Efficiently represent quantum states



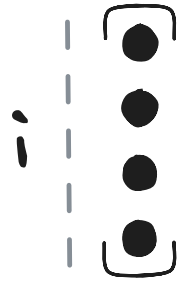
Machine learning applications



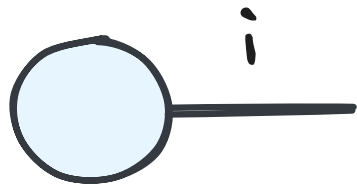
Solve high-dimensional linear systems

Study strongly correlated quantum systems

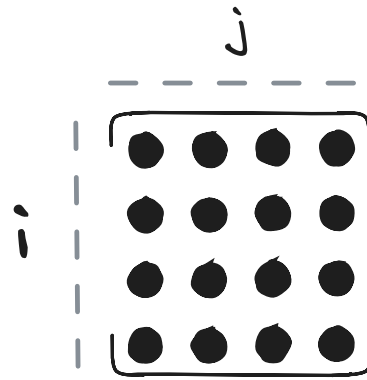
Vector



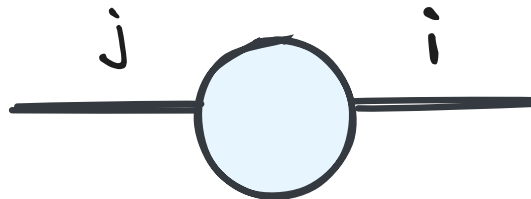
V_i



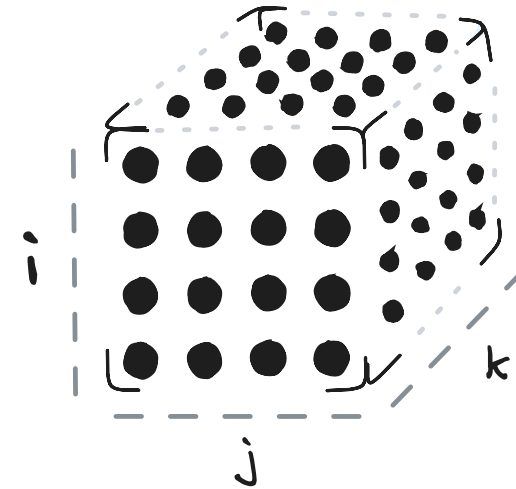
Matrix



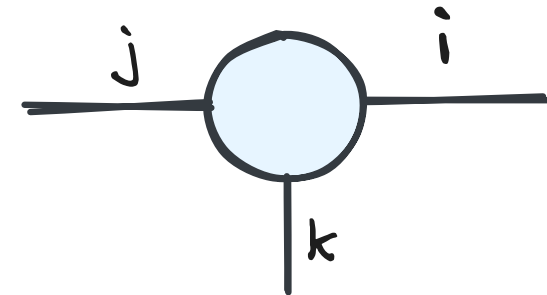
M_{ij}



3-order tensor

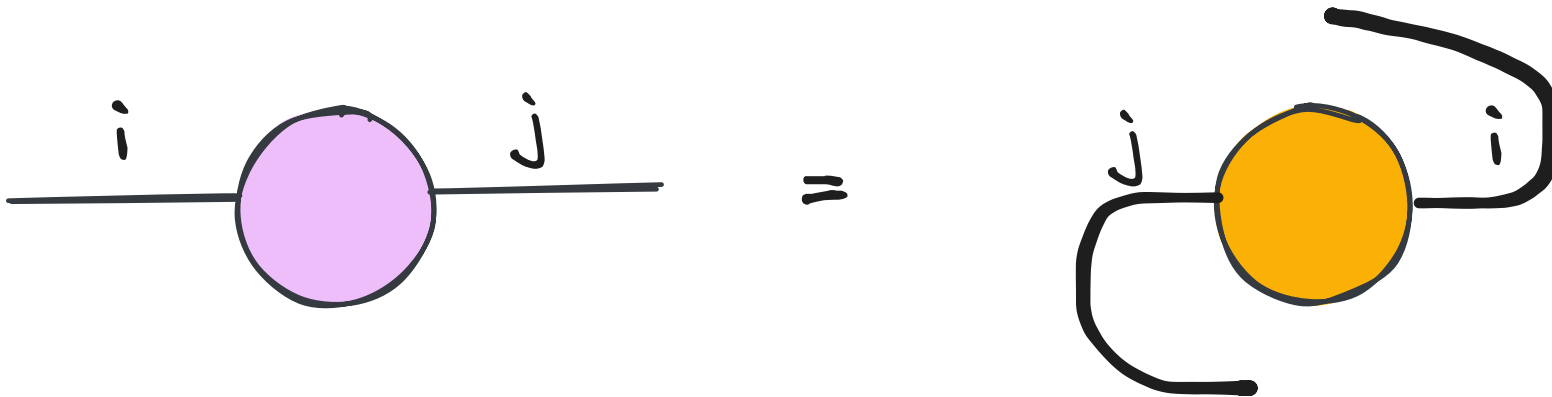


T_{ijk}

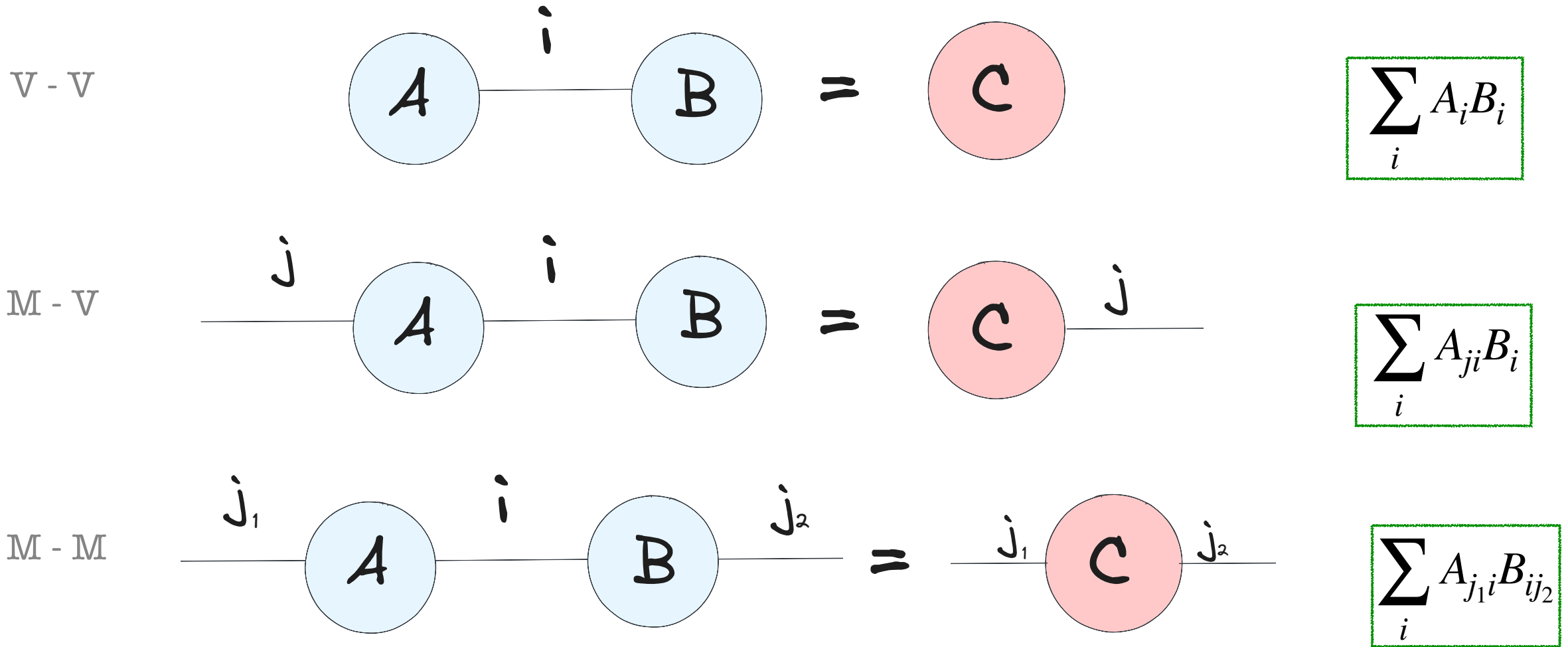


Transposition

$$A^T_{ji} = A_{ij}$$

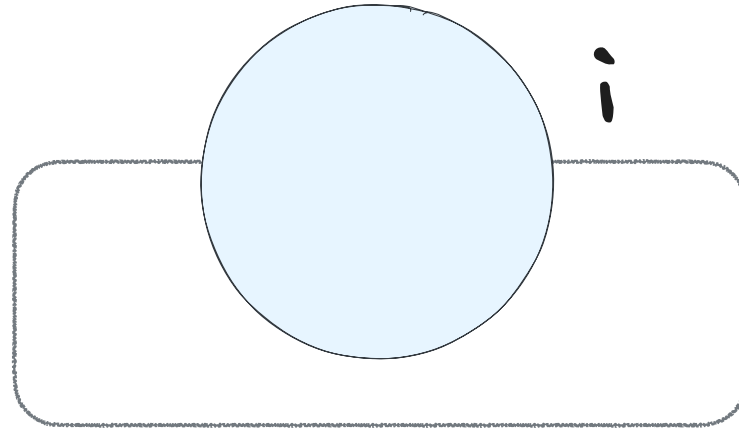


Contraction - summation over joint index



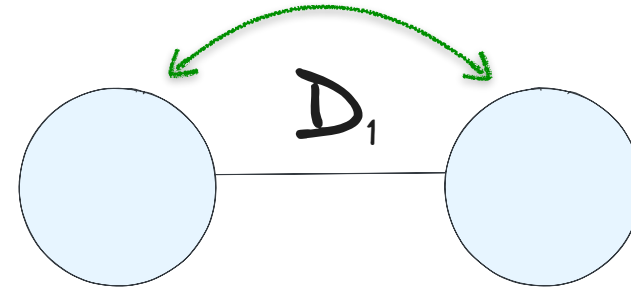
Trace of matrix

$$\text{tr}(A) = \sum_i A_{ii}$$

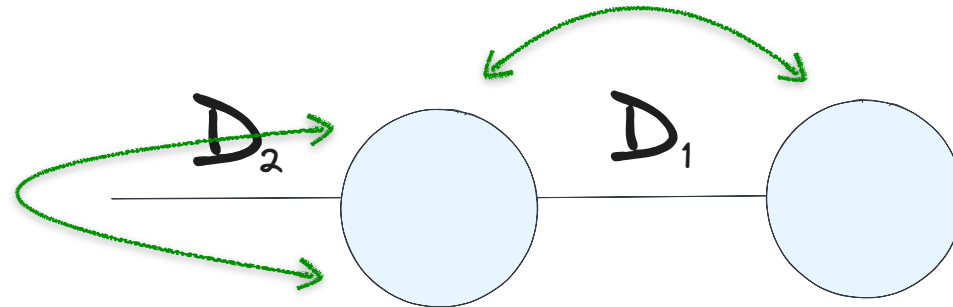


Contraction complexity

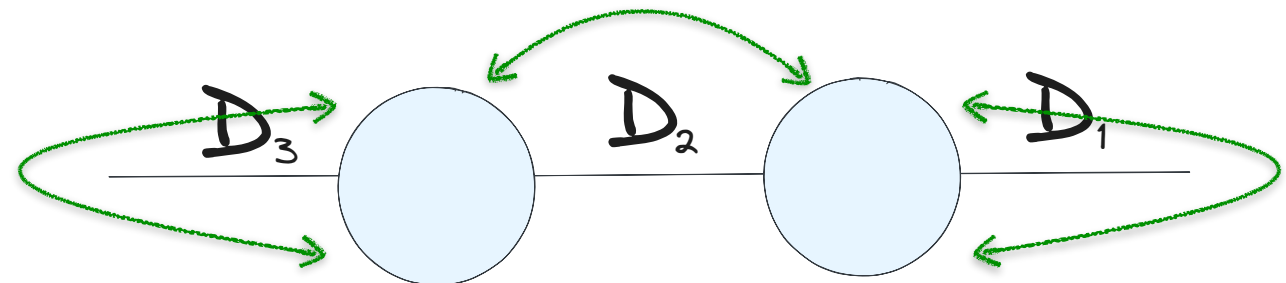
$$\mathcal{O}(D_1)$$



$$\mathcal{O}(D_1 D_2)$$

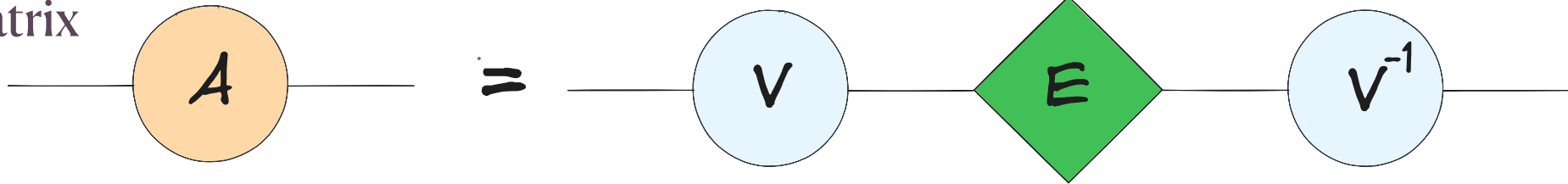


$$\mathcal{O}(D_1 D_2 D_3)$$



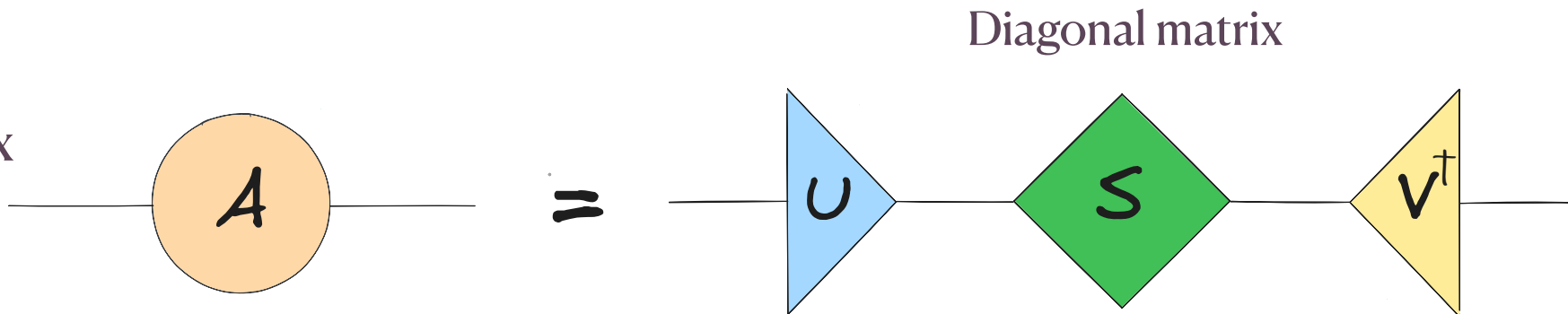
Eigendecomposition

Square matrix

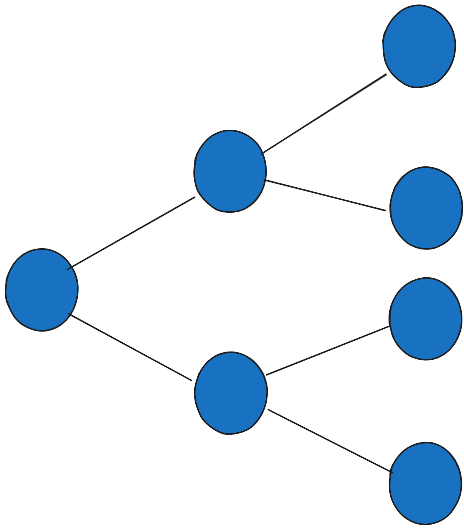


Singular Value Decomposition

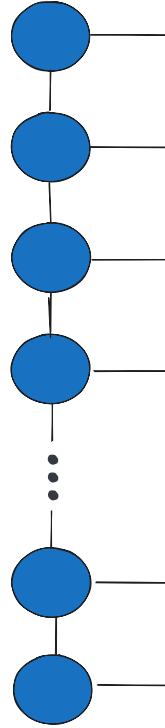
Any matrix



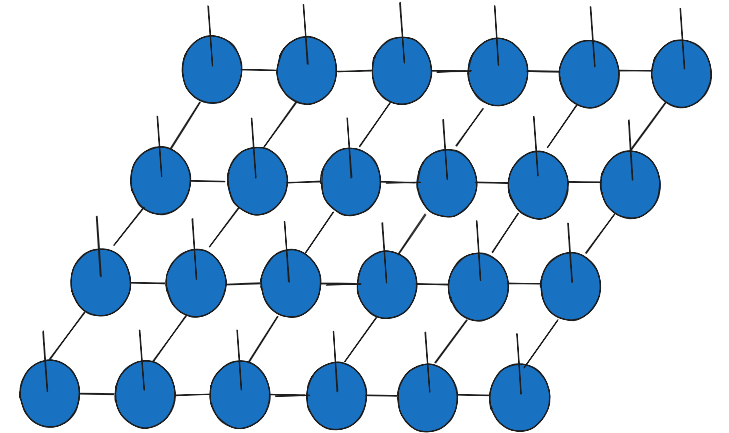
Tensor Networks



Tree



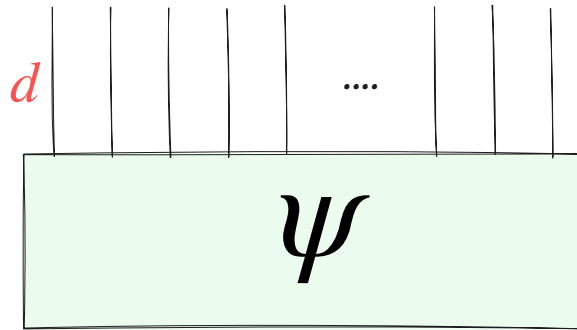
1D



2D

Quantum state

$$|\psi\rangle = \sum_{\{i_1, i_2 \dots i_N\}=0}^{d-1} C_{i_1, i_2 \dots i_N} | \overset{d}{\underbrace{i_1}}, \overset{d}{\underbrace{i_2}} \dots i_N \rangle$$

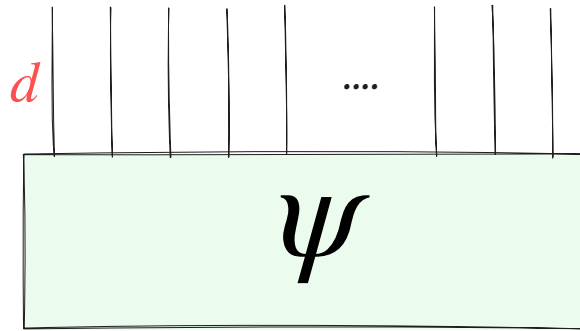


N-rank tensor of dimension d^N

$|i\rangle$ = basis state

d = Dimension of basis state

Quantum state



N-rank tensor of dimension d^N

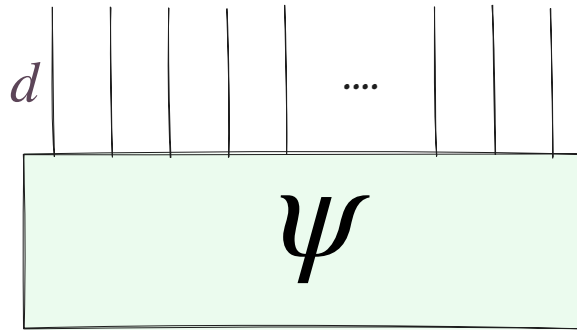
$$|\psi\rangle = \sum_{\{i_1, i_2 \dots i_N\}=0}^{d-1} \underbrace{C_{i_1, i_2 \dots i_N}} \overset{d}{|i_1\rangle}, \overset{d}{|i_2\rangle} \dots |i_N\rangle$$

$|i\rangle$ = basis state

d = Dimension of basis state

C = coefficient associated with basis state

Quantum state



N-rank tensor of dimension d^N

Dirac's Bra-Ket Notation

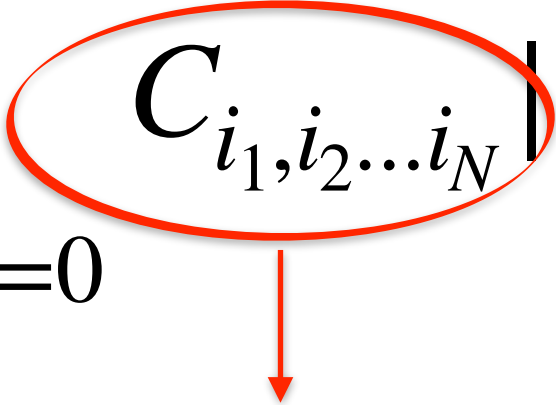
$$|\psi\rangle = \sum_{\{i_1, i_2 \dots i_N\}=0}^{d-1} C_{i_1, i_2 \dots i_N} |i_1, i_2 \dots i_N\rangle \text{ KET}$$

Vector in high-dim. vector space

$$\langle\psi| = \sum_{\{i_1, i_2 \dots i_N\}=0}^{d-1} C_{i_1, i_2 \dots i_N}^* \langle i_1, i_2 \dots i_N| \text{ BRA}$$

Dual vector in high-dim. vector space

Quantum state

$$|\psi\rangle = \sum_{\{i_1, i_2 \dots i_N\} = 0}^{d-1} C_{i_1, i_2 \dots i_N} |i_1, i_2 \dots i_N\rangle$$


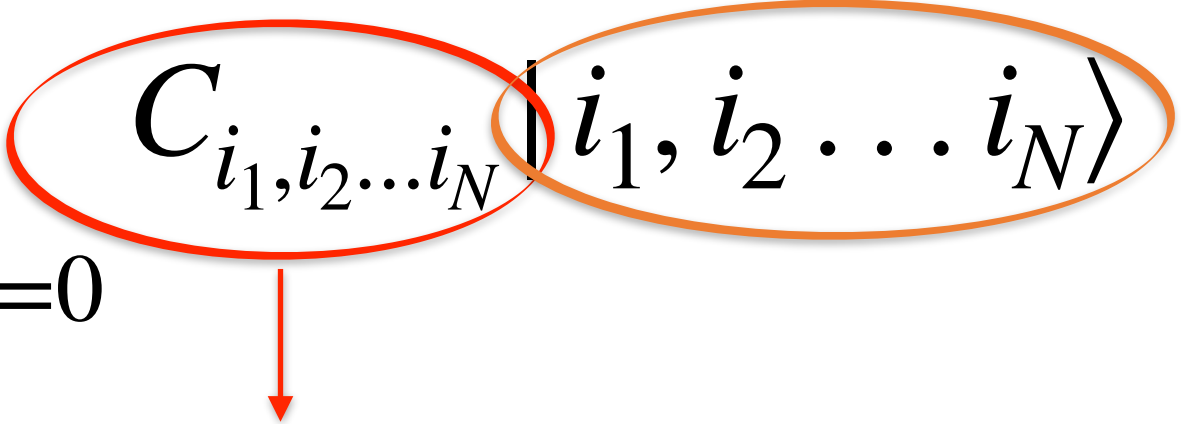
Represent as Tensor Network

Quantum state

$$|\psi\rangle = \sum_{\{i_1, i_2 \dots i_N\}=0}^{d-1} C_{i_1, i_2 \dots i_N} |i_1, i_2 \dots i_N\rangle$$

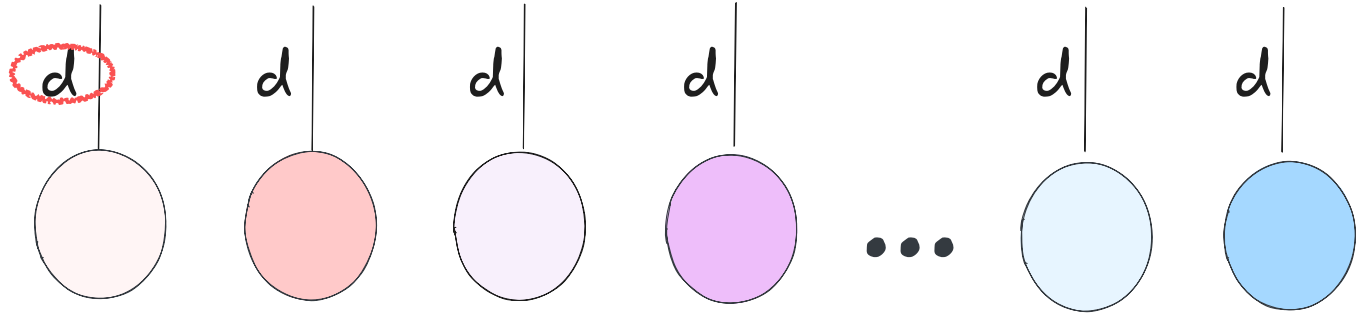
Product of local tensors

Represent as Tensor Network

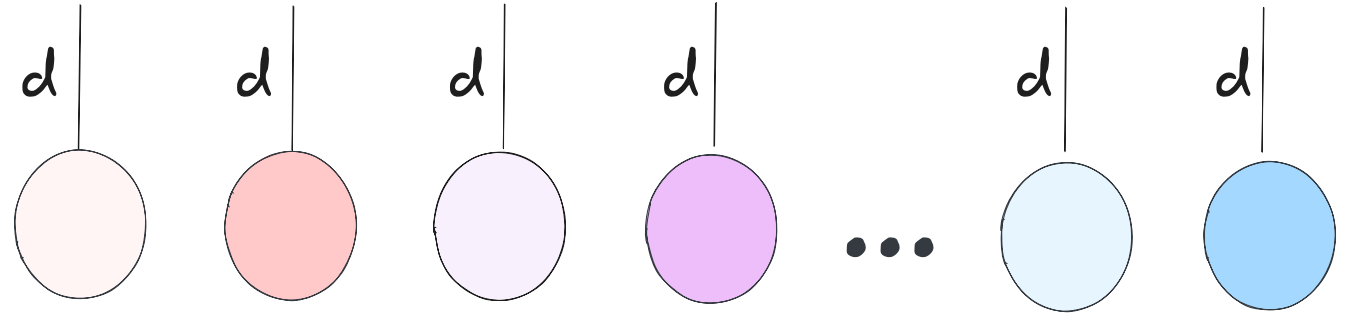


Product State

physical dimension

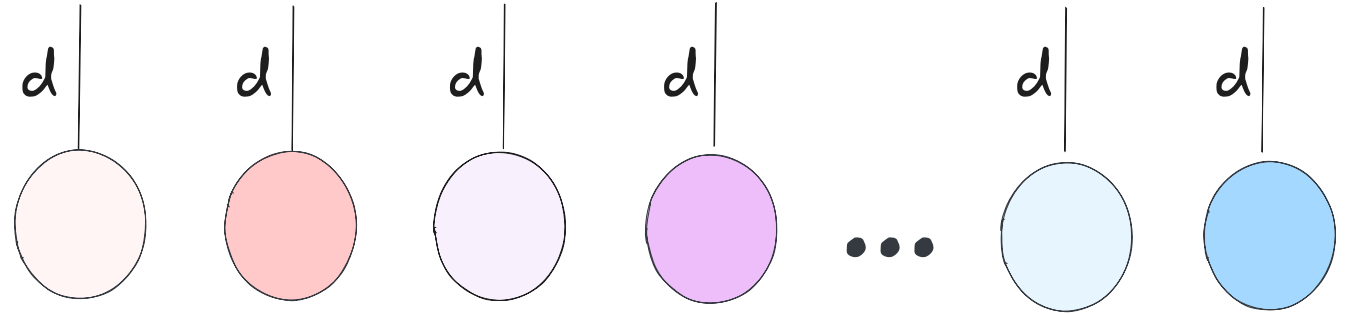


Product State



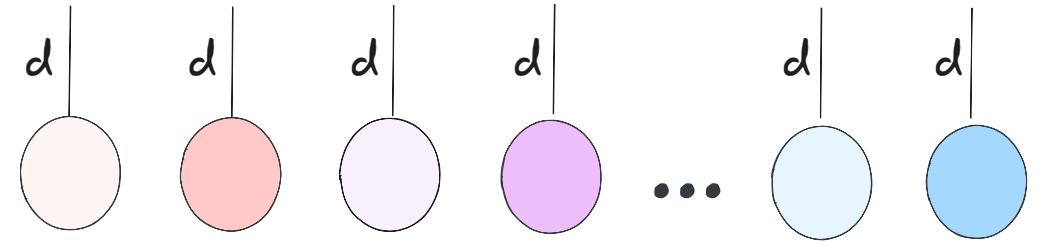
$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_N\rangle$$

Product State



$$\begin{aligned} |\psi\rangle &= |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_N\rangle \\ &= \left(\sum_{i_1} c_1^{i_1} |i_1\rangle \right) \otimes \dots \otimes \left(\sum_{i_N} c_N^{i_N} |i_N\rangle \right) \end{aligned}$$

Product State



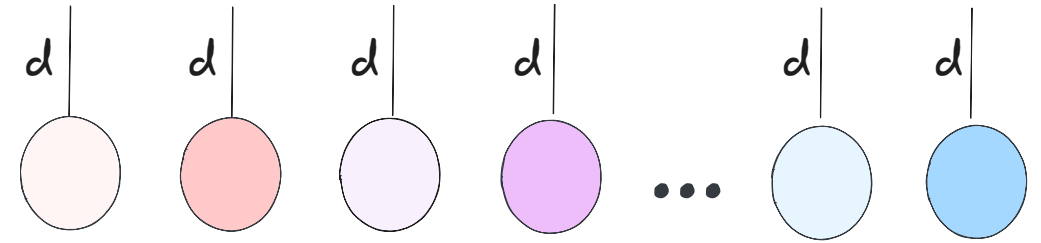
$$d = 2$$

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_N\rangle$$

$$= \left(\sum_{i_1} c_1^{i_1} |i_1\rangle \right) \otimes \dots \otimes \left(\sum_{i_N} c_N^{i_N} |i_N\rangle \right)$$

$$|\Psi\rangle = \left(0 \times \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \otimes$$

Product State



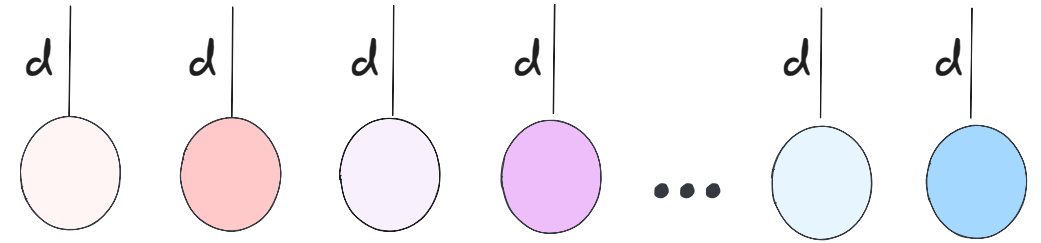
$$d = 2$$

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_N\rangle$$

$$= \left(\sum_{i_1} c_1^{i_1} |i_1\rangle \right) \otimes \dots \otimes \left(\sum_{i_N} c_N^{i_N} |i_N\rangle \right)$$

$$|\psi\rangle = \left(0 \times \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \otimes$$

Product State



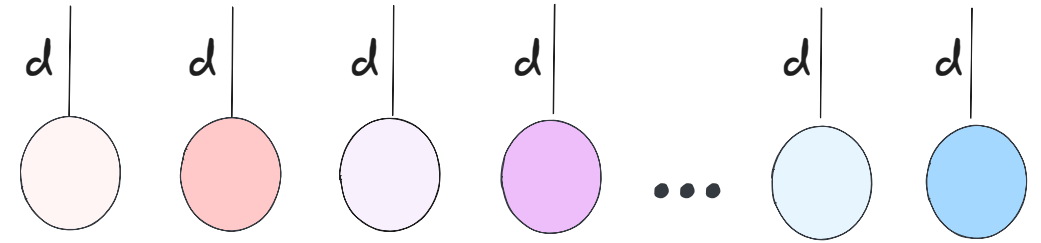
$$d = 2$$

$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_N\rangle$$

$$= \left(\sum_{i_1} c_1^{i_1} |i_1\rangle \right) \otimes \dots \otimes \left(\sum_{i_N} c_N^{i_N} |i_N\rangle \right)$$

$$|\Psi\rangle = \left(0 \times \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \otimes \left(1 \times \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

Product State



$$d = 2$$

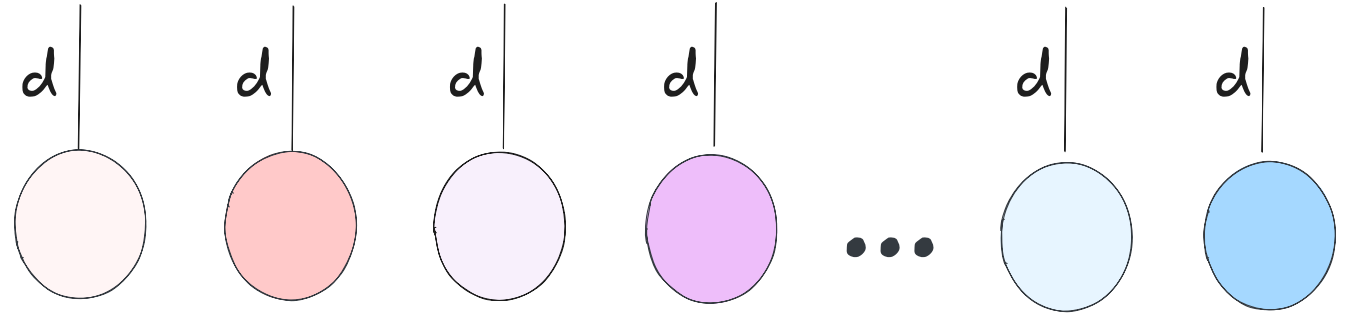
$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_N\rangle$$

$$= \left(\sum_{i_1} c_1^{i_1} |i_1\rangle \right) \otimes \dots \otimes \left(\sum_{i_N} c_N^{i_N} |i_N\rangle \right)$$

$$|\Psi\rangle = \left(0 \times \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 1 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right) \otimes \left(1 \times \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0 \times \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

$$|\Psi\rangle = |1\rangle \otimes |0\rangle$$

Product State



$$|\psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle \otimes \dots \otimes |\psi_N\rangle$$

$$= \left(\sum_{i_1} c_1^{i_1} |i_1\rangle \right) \otimes \dots \otimes \left(\sum_{i_N} c_N^{i_N} |i_N\rangle \right) = \sum_{\{i_1, i_2, \dots, i_N\}}^{d-1} c_1^{i_1} c_2^{i_2} \dots c_N^{i_N} |i_1, i_2, \dots, i_N\rangle$$

product of scalars

Bell state

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

entangled quantum state

Bell state

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \quad \text{entangled quantum state}$$

$|00\rangle$
 $|01\rangle$
 $|10\rangle$
 $|11\rangle$

Tensor product combinations of 2 basis states

Bell state

$$|\Psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

entangled quantum state

Coeffs basis states

1	$\times$	$ 00\rangle$
0	$\times$	$ 01\rangle$
0	$\times$	$ 10\rangle$
1	$\times$	$ 11\rangle$

Coefficient of Bell State

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

coefficient matrix

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \xrightarrow{\text{reshape}} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

reshape

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

coefficient matrix

$1 \times 100 \rangle$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

reshape

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

coefficient matrix

$$1 \times |00\rangle$$

$$0 \times |01\rangle$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

reshape

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

coefficient matrix

$$1 \times |00\rangle$$

$$0 \times |01\rangle$$

$$0 \times |10\rangle$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

reshape

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

coefficient matrix

$$1 \times |00\rangle$$

$$0 \times |01\rangle$$

$$0 \times |10\rangle$$

$$1 \times |11\rangle$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

reshape

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

coefficient matrix

1	x	100>
0	x	101>
0	x	110>
1	x	111>

basis states

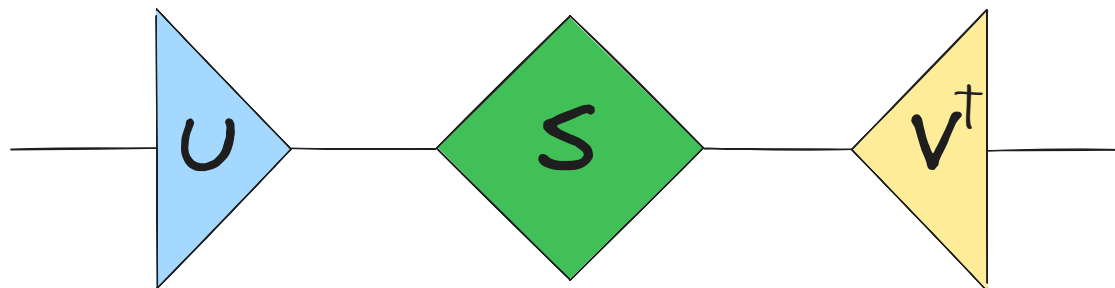
$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} \longrightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \longrightarrow \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

Bell state

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

SVD
→

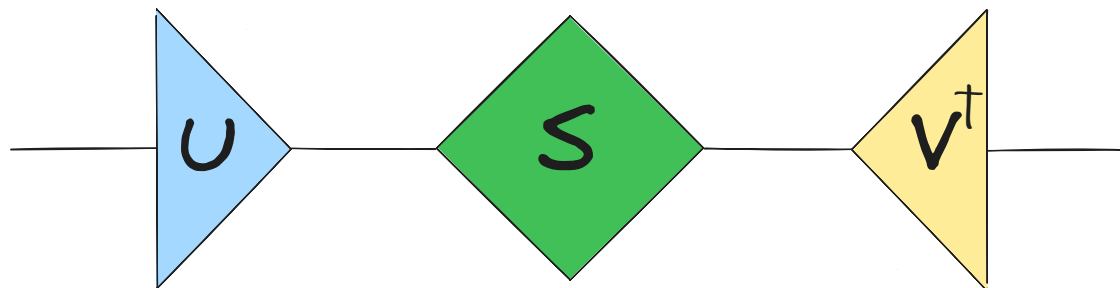
$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$



$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

SVD
→

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

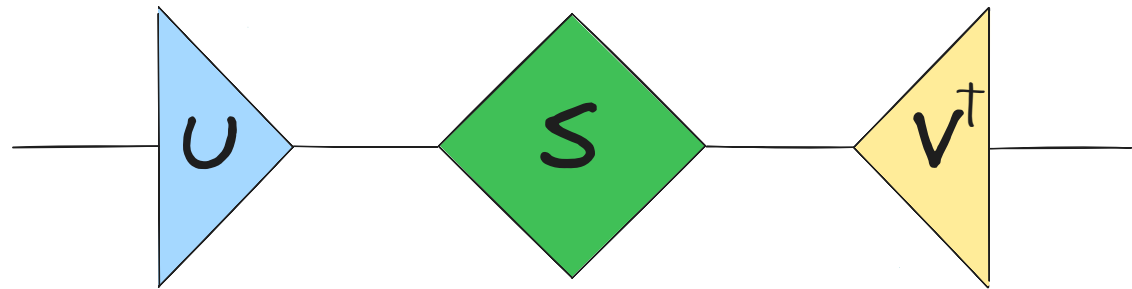


$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

SVD
→

singular values

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} \textcircled{1} & 0 \\ 0 & \textcircled{1} \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$



$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \xrightarrow{\text{SVD}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} \textcircled{1} & 0 \\ 0 & \textcircled{1} \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

singular values

$$\lambda_i = \frac{1}{\sqrt{2}}$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \xrightarrow{\text{SVD}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} \textcircled{1} & 0 \\ 0 & \textcircled{1} \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

singular values

$$\lambda_i = \frac{1}{\sqrt{2}} \longrightarrow D = 2$$

Schmidt rank

(number of non-zero singular values)

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \xrightarrow{\text{SVD}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$\xleftrightarrow{d=2}$ (horizontal arrow above the third matrix) and $\updownarrow d=2$ (vertical arrow to the right of the third matrix)

$$\lambda_i = \frac{1}{\sqrt{2}} \longrightarrow \boxed{D \leq \min(d_1, d_2)}$$

Schmidt rank

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \xrightarrow{\text{SVD}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$\overleftrightarrow{d=2}$
 $\updownarrow d=2$

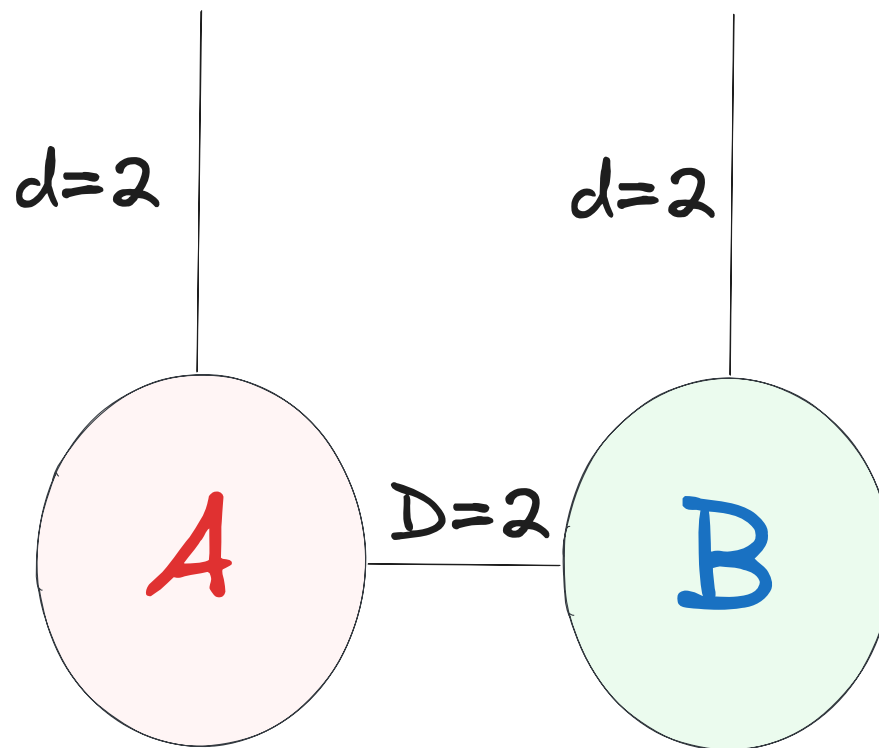
$$\lambda_i = \frac{1}{\sqrt{2}} \longrightarrow \boxed{D \leq \min(d_1, d_2)}$$

Schmidt rank

max entanglement
of quantum state

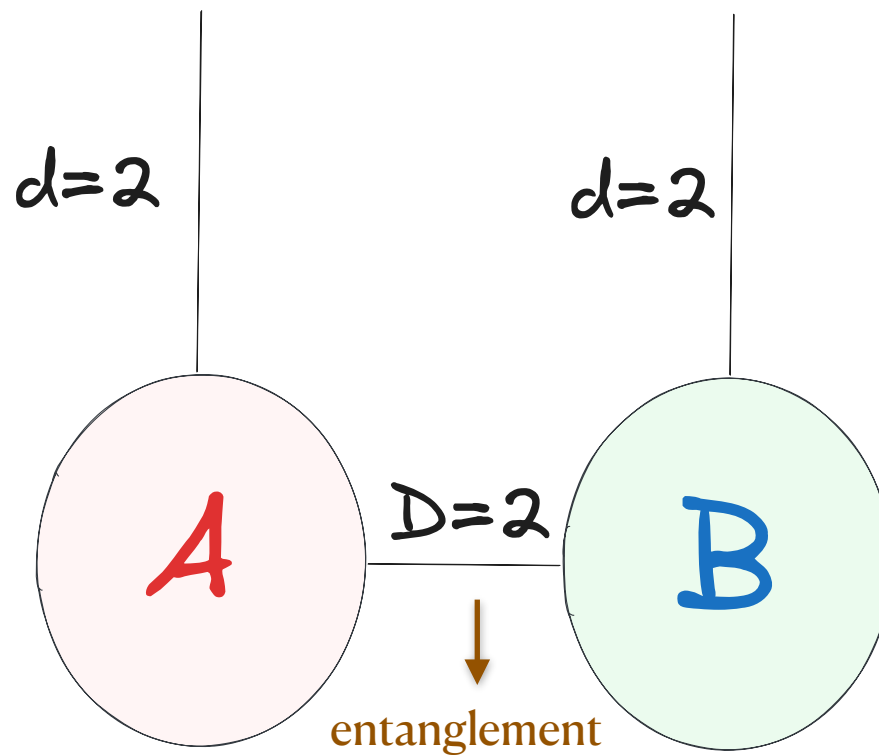
- Same singular values
- Schmidt rank = 2

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}$$



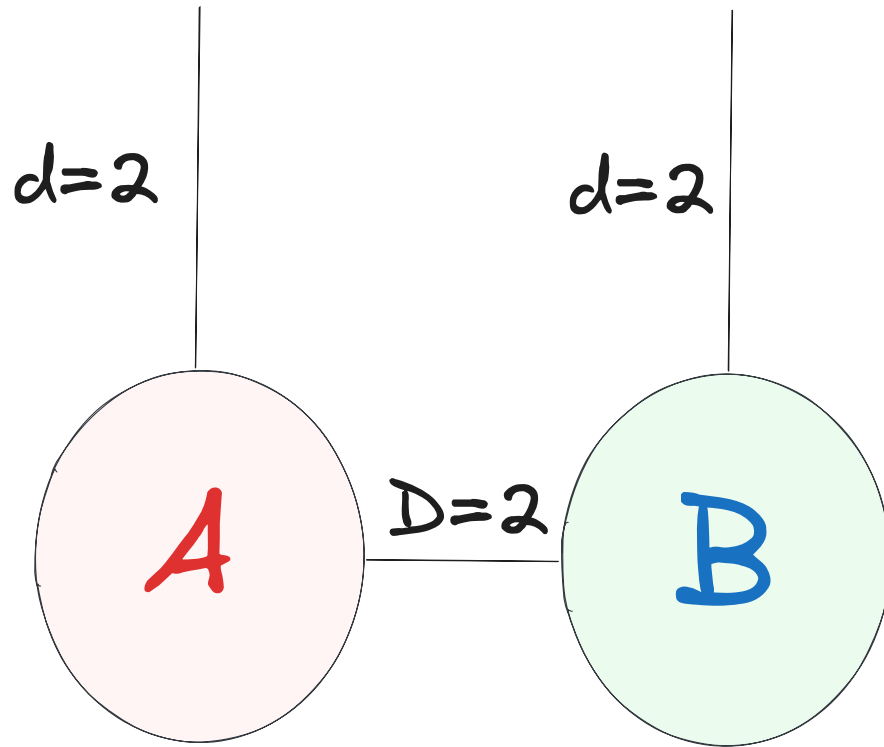
Matrix Product State

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \times \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}$$



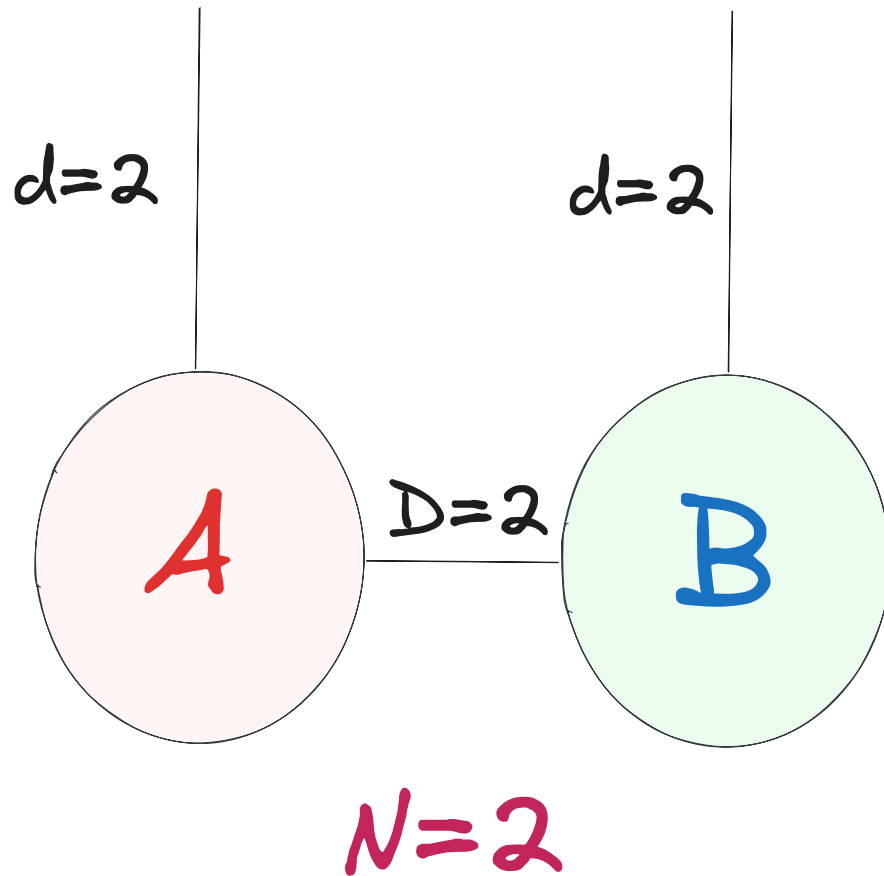
Matrix Product State

Matrix Product State



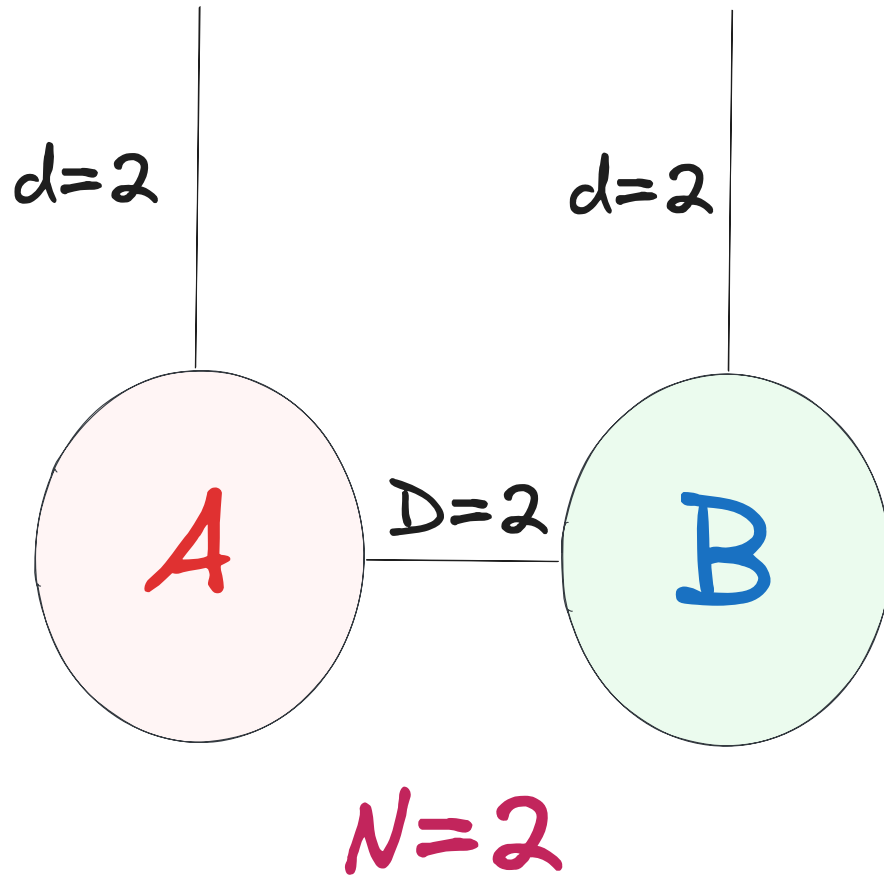
Number of parameters
 $\mathcal{O}(NdD^2)$

Matrix Product State



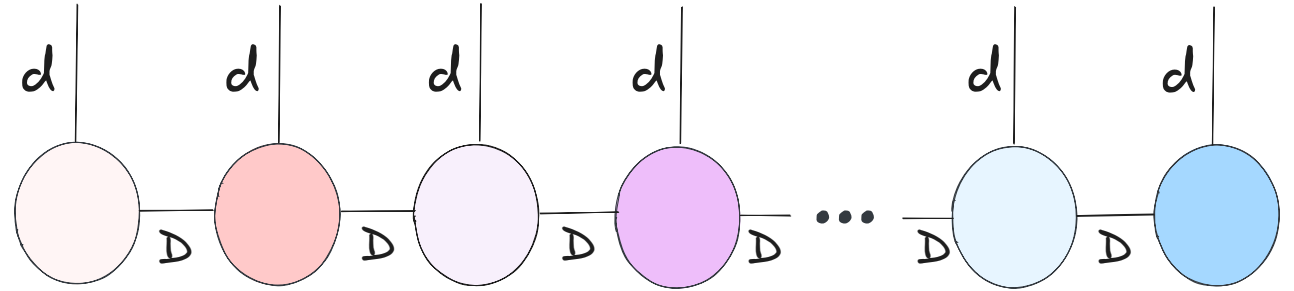
Number of parameters
 $\mathcal{O}(NdD^2)$

Matrix Product State



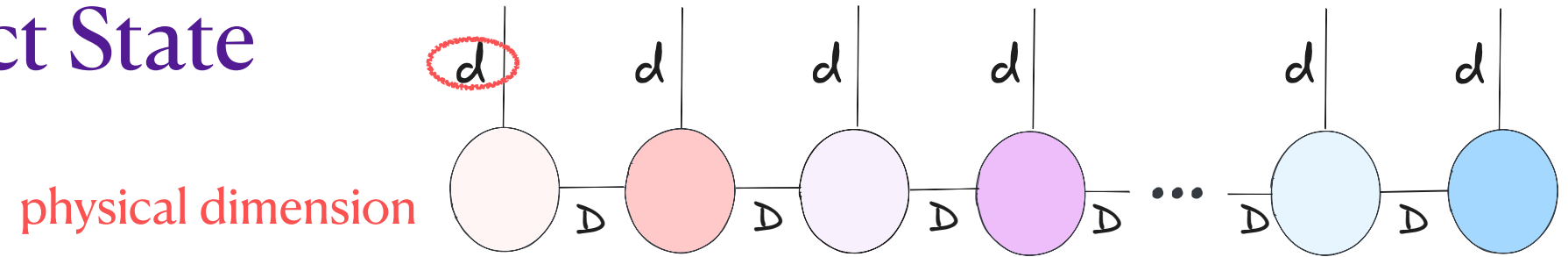
Number of parameters
16

Matrix Product State



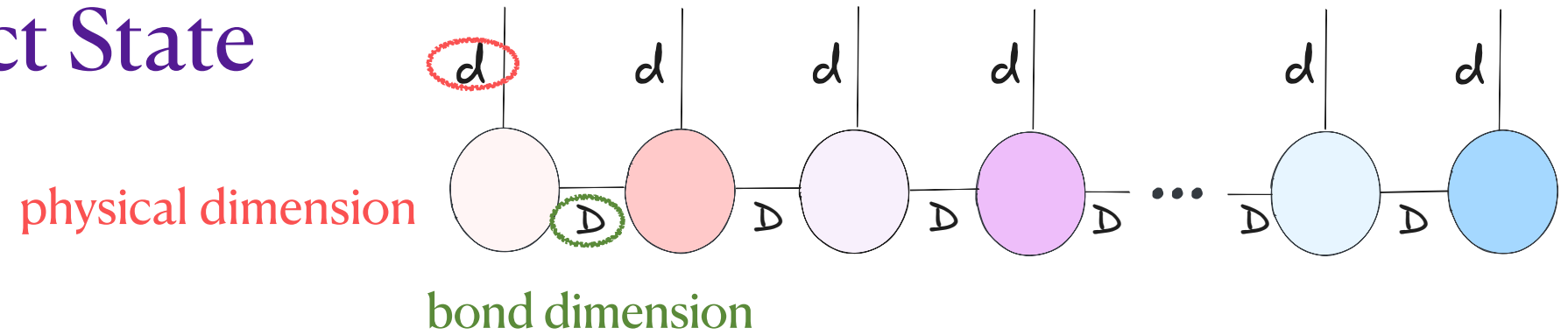
$$|\psi\rangle = \sum_{\{i_1, i_2 \dots i_N\}=0}^{d-1} C_{i_1, i_2 \dots i_N} |i_1, i_2 \dots i_N\rangle$$

Matrix Product State



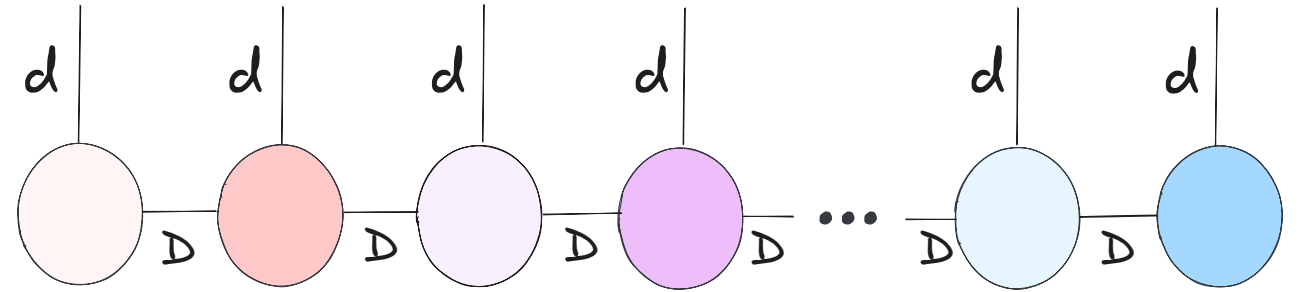
$$|\psi\rangle = \sum_{\{i_1, i_2 \dots i_N\}=0}^{d-1} C_{i_1, i_2 \dots i_N} |i_1, i_2 \dots i_N\rangle$$

Matrix Product State



$$|\psi\rangle = \sum_{\{i_1, i_2 \dots i_N\}}^{d-1} C_{i_1, i_2 \dots i_N} |i_1, i_2 \dots i_N\rangle$$

Matrix Product State

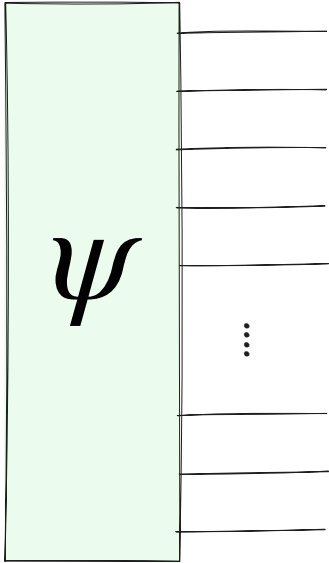


$$|\psi\rangle = \sum_{\{i_1, i_2 \dots i_N\}=0}^{d-1} C_{i_1, i_2 \dots i_N} |i_1, i_2 \dots i_N\rangle$$

product of matrices

Matrix Product State

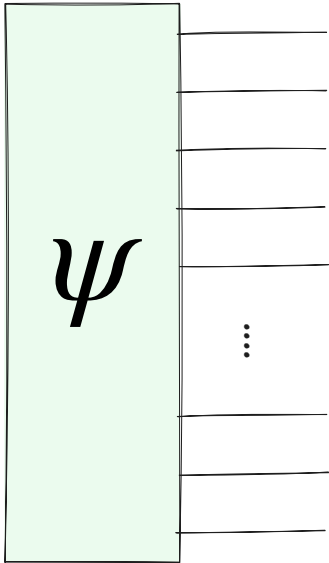
N-rank tensor



$$\# \text{ params} = d^N$$

Matrix Product State

N-rank tensor



Decomposed w/

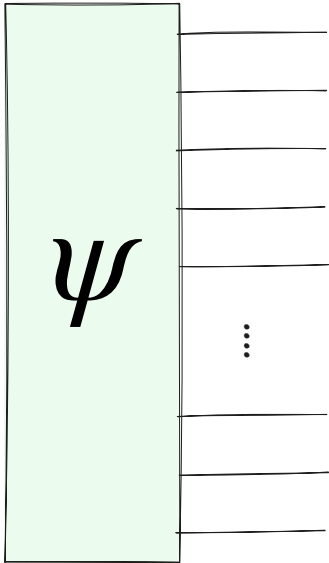
Sequential
SVDs



$$\# \text{ params} = d^N$$

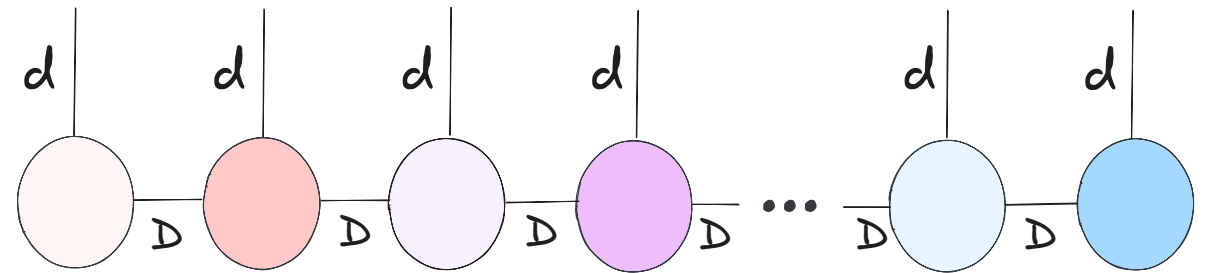
Matrix Product State

N-rank tensor



Decomposed w/

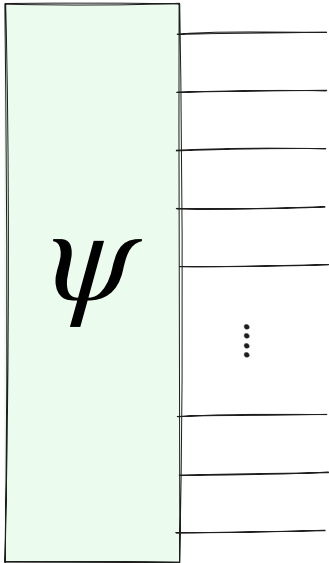
Sequential
SVDs



$$\# \text{ params} = d^N$$

Matrix Product State

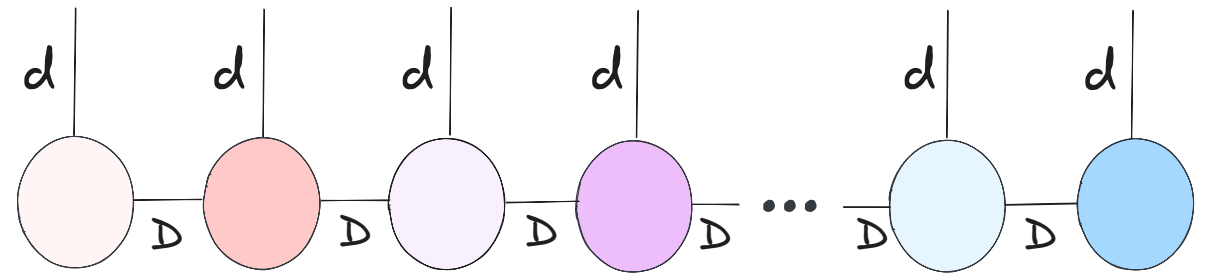
N-rank tensor



$$\# \text{ params} = d^N$$

Decomposed w/

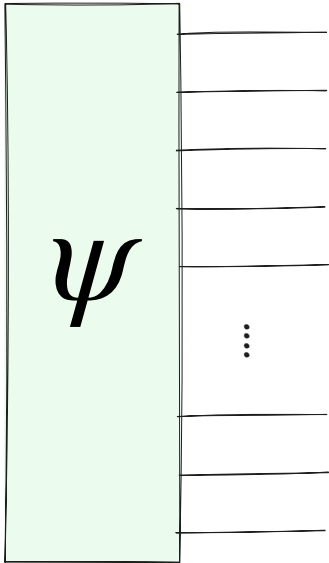
Sequential
SVDs



$$\# \text{ params} = NdD^2$$

Matrix Product State

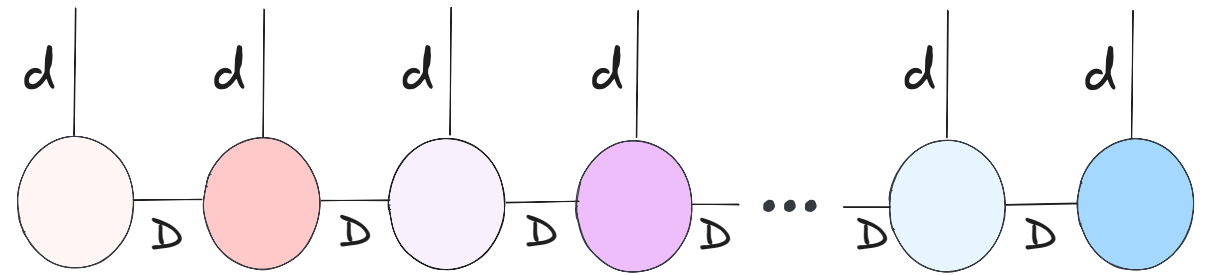
N-rank tensor



$$\# \text{ params} = d^N$$

Decomposed w/

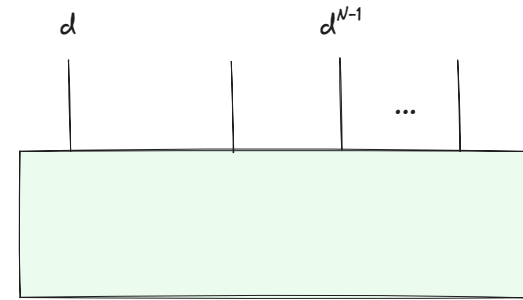
Sequential
SVDs



$$\# \text{ params} = NdD^2$$

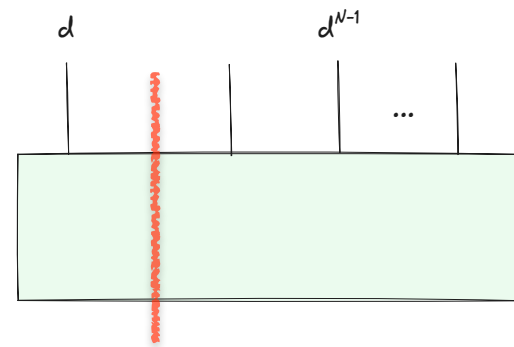
Singular Value Decomposition

$$d \times d^{N-1}$$



Singular Value Decomposition

$$d \times d^{N-1}$$

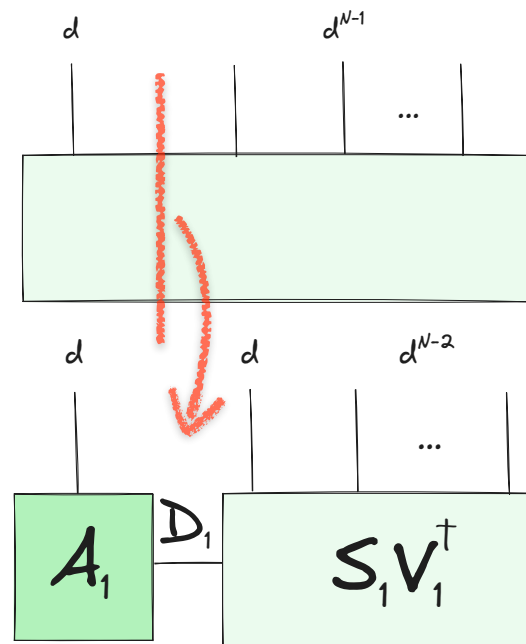


Singular Value Decomposition

$$d \times d^{N-1}$$



$$A_1 S_1 V_1^\dagger$$

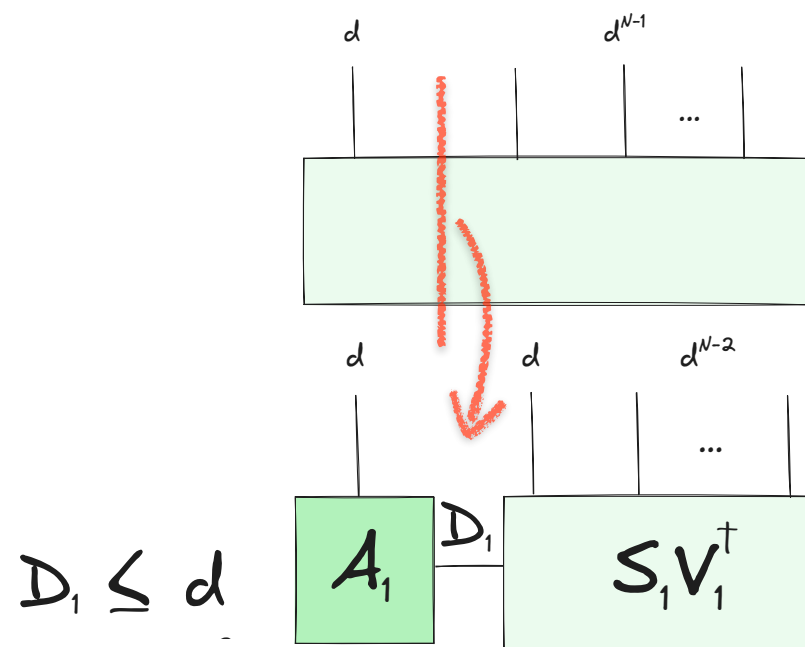


Singular Value Decomposition

$$d \times d^{N-1}$$

↓

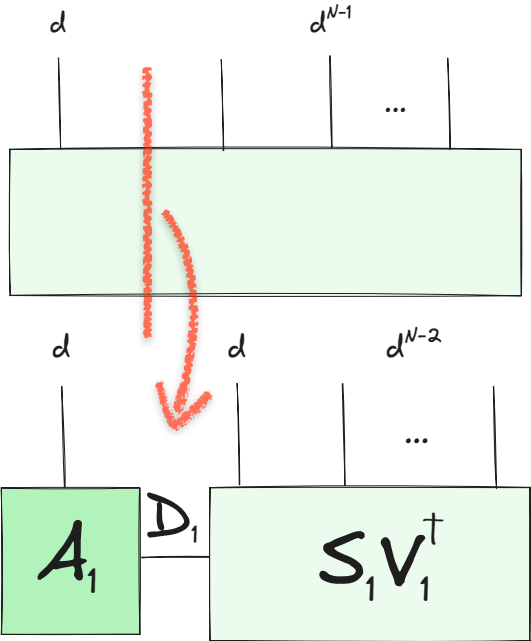
$$d \times D_1 \quad A_1 \quad S_1 V_1^\dagger$$



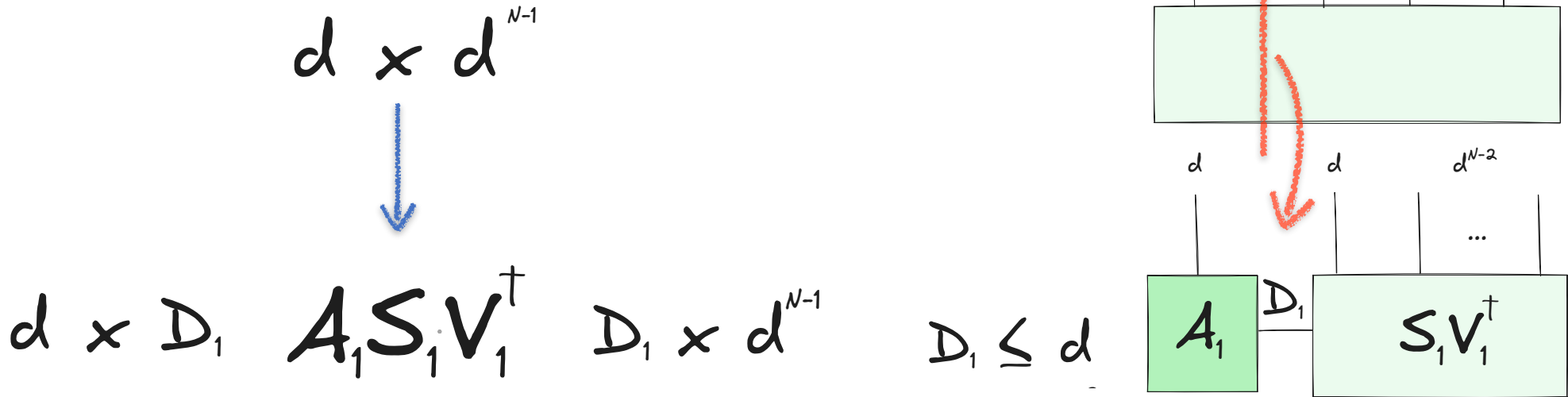
Singular Value Decomposition

$$\begin{array}{c}
 d \times d^{N-1} \\
 \downarrow \\
 d \times D_1 \quad A_1 \textcircled{S_1 V_1^\dagger} D_1 \times d^{N-1}
 \end{array}$$

$$D_1 \leq d$$



Singular Value Decomposition



Singular Value Decomposition

$$d \times d^{N-1}$$

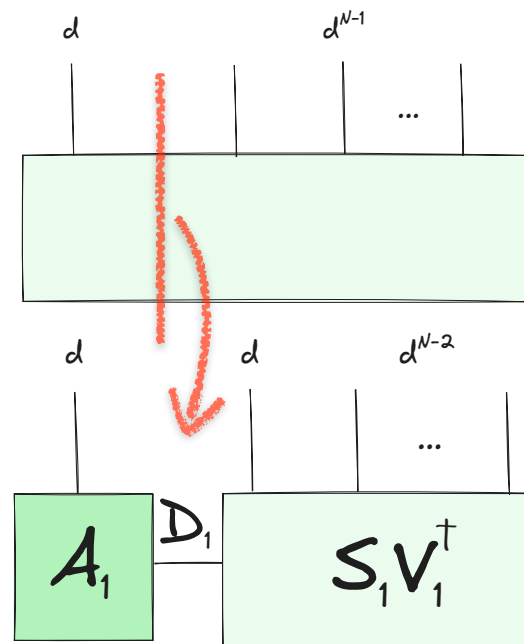
↓

$$d \times D_1 \quad A_1 \quad \boxed{S_1 V_1^T} \quad D_1 \times d^{N-1}$$

↘

$$D_1 d \times d^{N-2}$$

$$D_1 \leq d$$



Singular Value Decomposition

$$d \times d^{N-1}$$

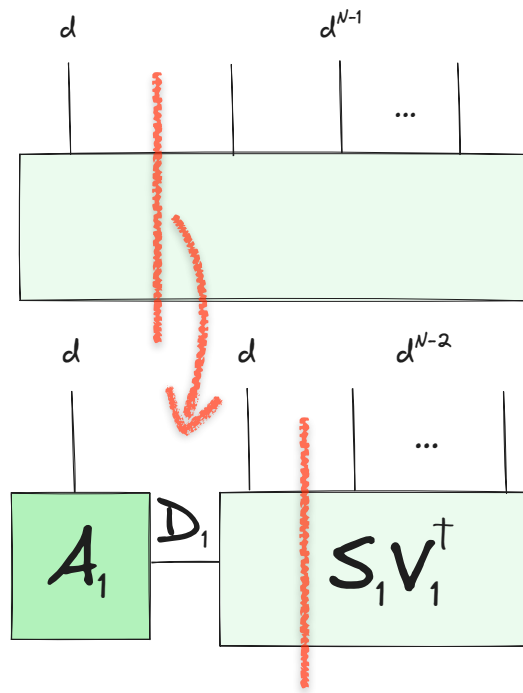
↓

$$d \times D_1 \quad A_1 \quad \boxed{S_1 V_1^T} \quad D_1 \times d^{N-1}$$

↘

$$D_1 d \times d^{N-2}$$

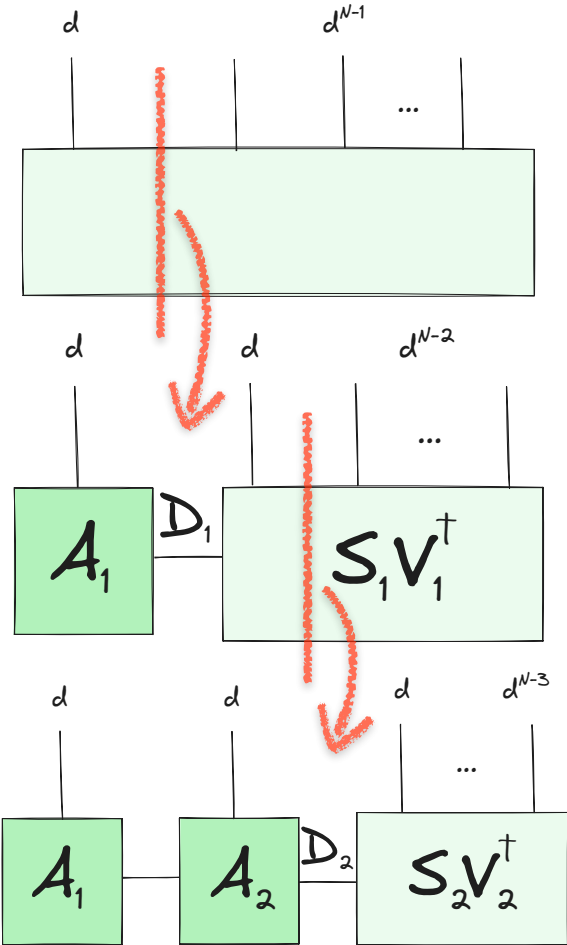
$$D_1 \leq d$$



Singular Value Decomposition

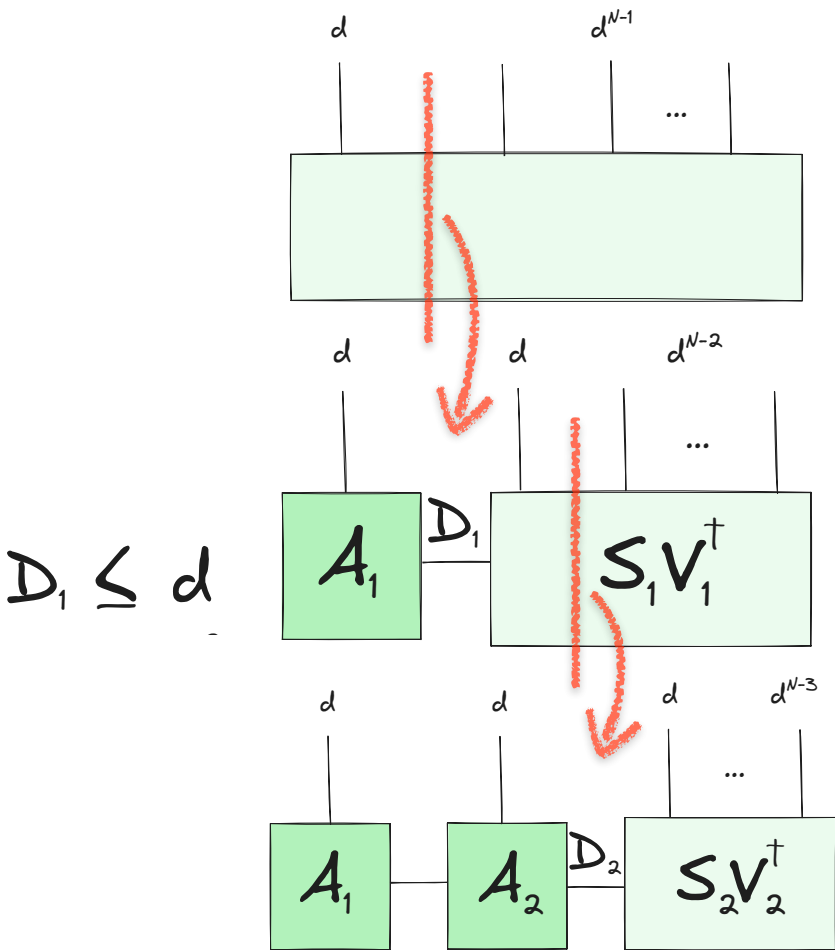
$$\begin{aligned}
 & d \times d^{N-1} \\
 & \downarrow \\
 & d \times D_1 \quad A_1 S_1 V_1^\dagger \quad D_1 \times d^{N-1} \\
 & \qquad \qquad \qquad \searrow \\
 & \qquad \qquad \qquad D_1 d \times d^{N-2}
 \end{aligned}$$

$$D_1 \leq d$$



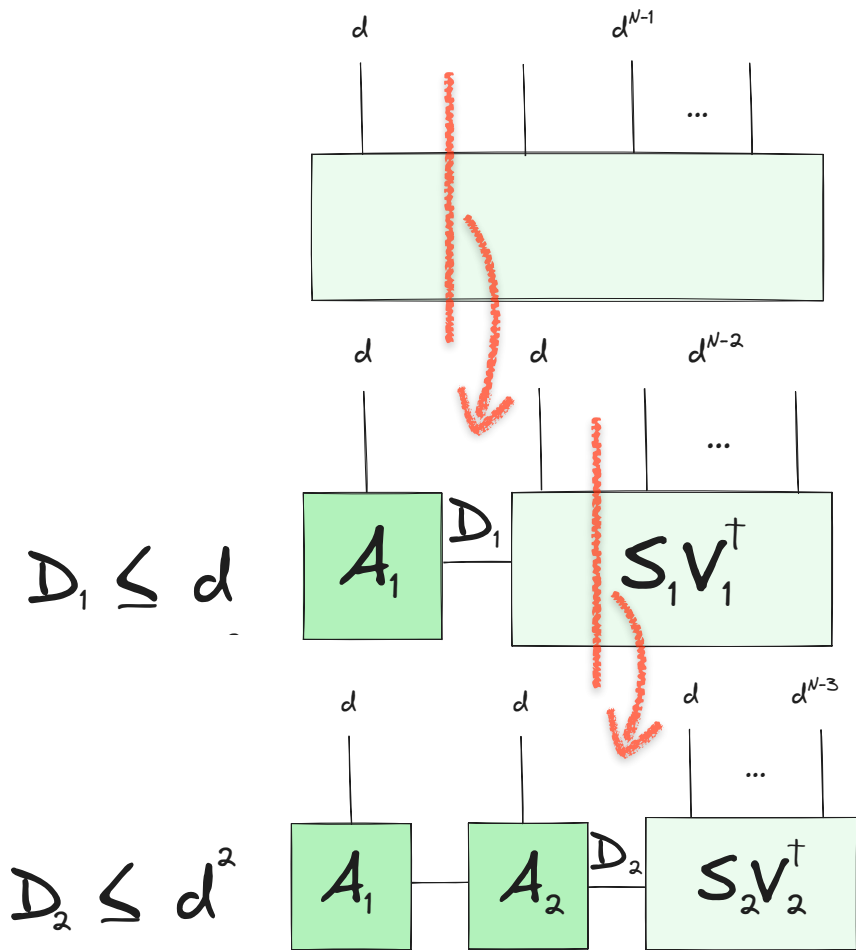
Singular Value Decomposition

$$\begin{aligned}
 & d \times d^{N-1} \\
 & \downarrow \\
 & d \times D_1 \quad A_1 S_1 V_1^\dagger \quad D_1 \times d^{N-1} \\
 & \quad \quad \quad \searrow \\
 & \quad \quad \quad D_1 d \times d^{N-2} \\
 & \downarrow \\
 & A_1 A_2 S_2 V_2^\dagger
 \end{aligned}$$



Singular Value Decomposition

$$\begin{aligned}
 & d \times d^{N-1} \\
 & \downarrow \\
 & d \times D_1 \quad A_1 S_1 V_1^\dagger \quad D_1 \times d^{N-1} \\
 & \quad \quad \quad \searrow \\
 & \quad \quad \quad D_1 d \times d^{N-2} \\
 & \downarrow \\
 & A_1 A_2 S_2 V_2^\dagger
 \end{aligned}$$



Singular Value Decomposition

$$d \times d^{N-1}$$

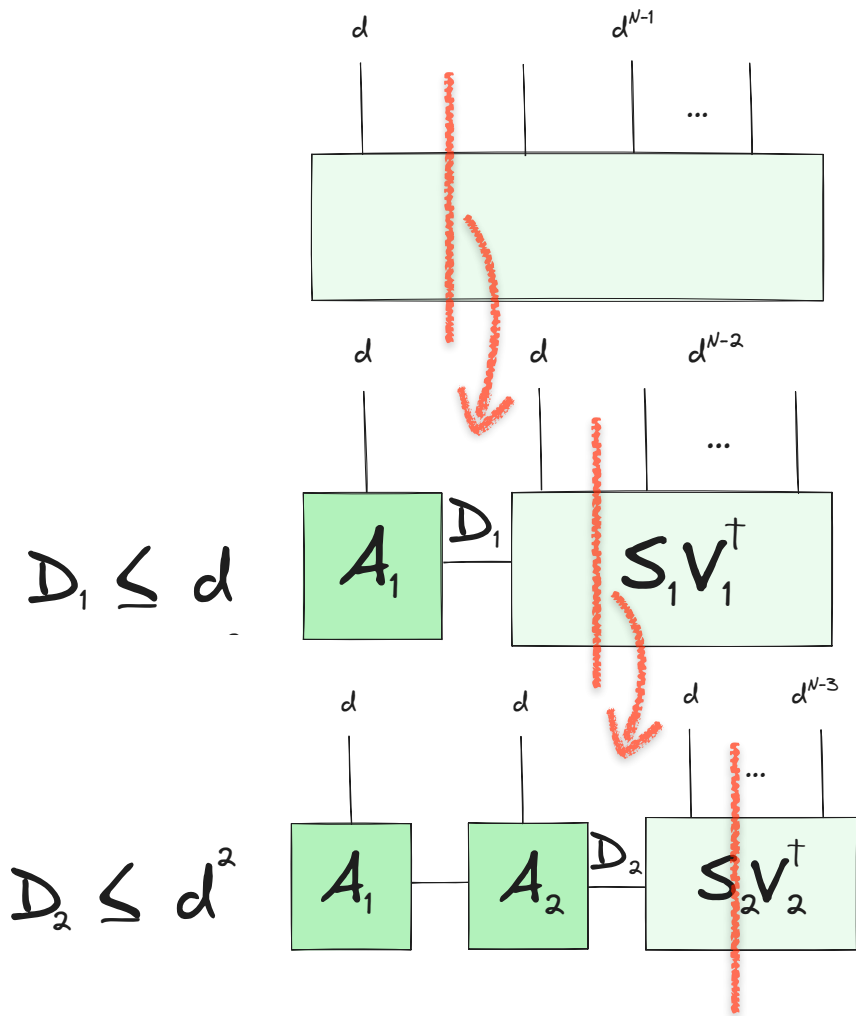
$$\downarrow$$

$$d \times D_1 \quad A_1 S_1 V_1^\dagger \quad D_1 \times d^{N-1}$$

$$\downarrow$$

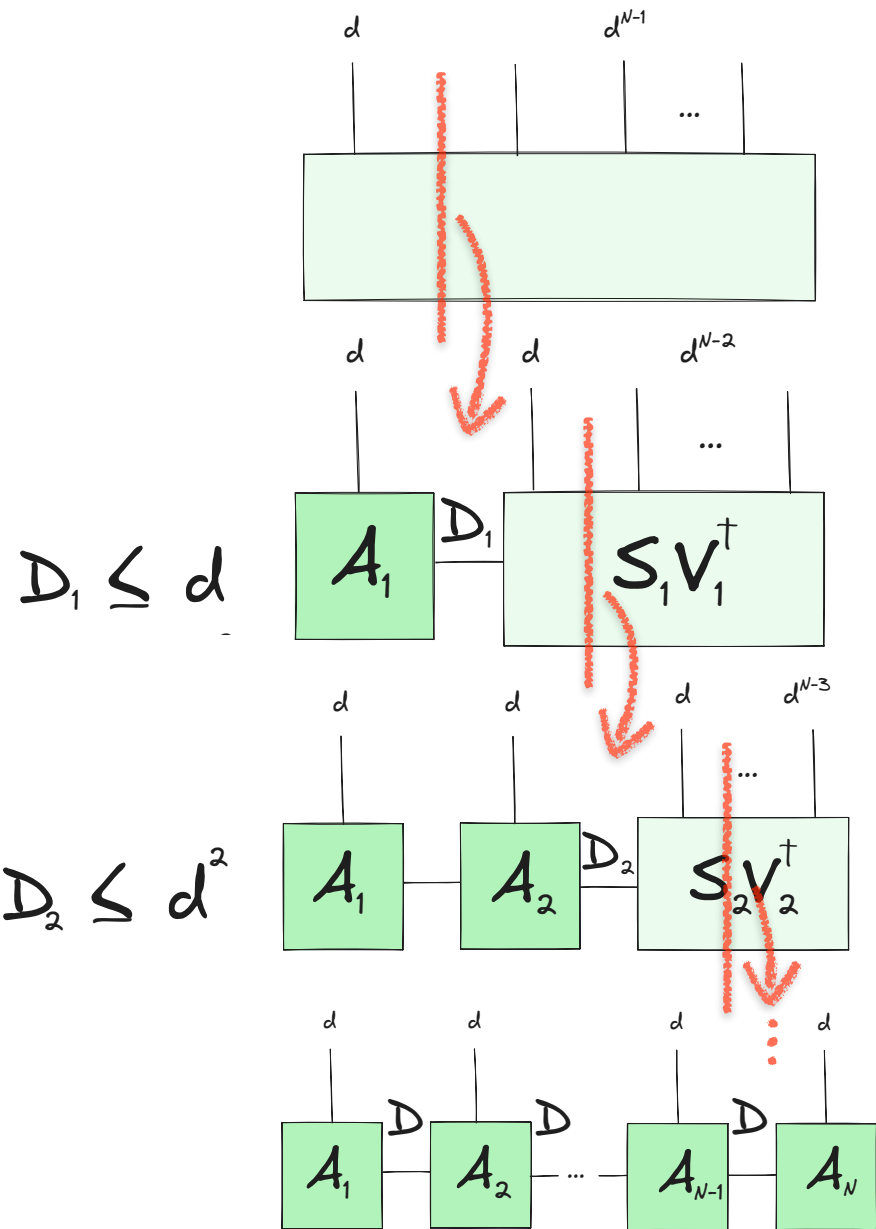
$$A_1 A_2 S_2 V_2^\dagger$$

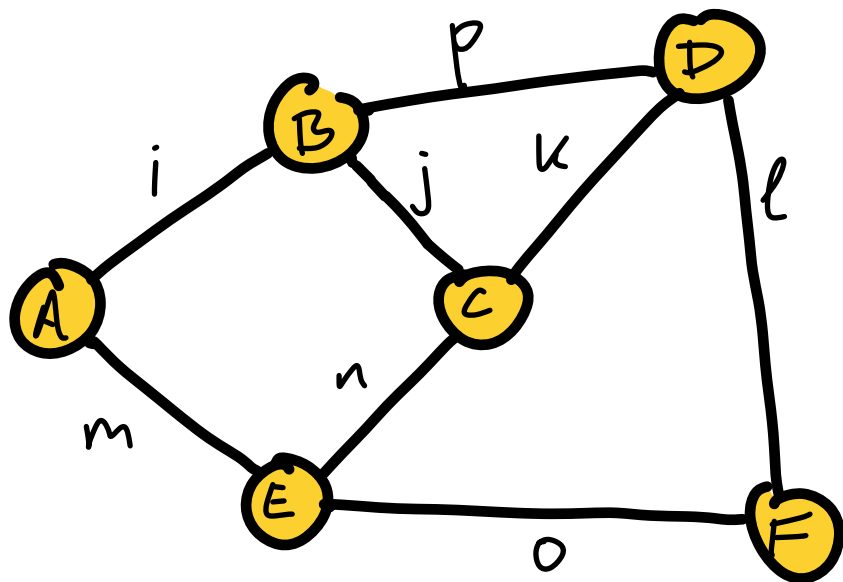
$$\nearrow D_1 d \times d^{N-2}$$



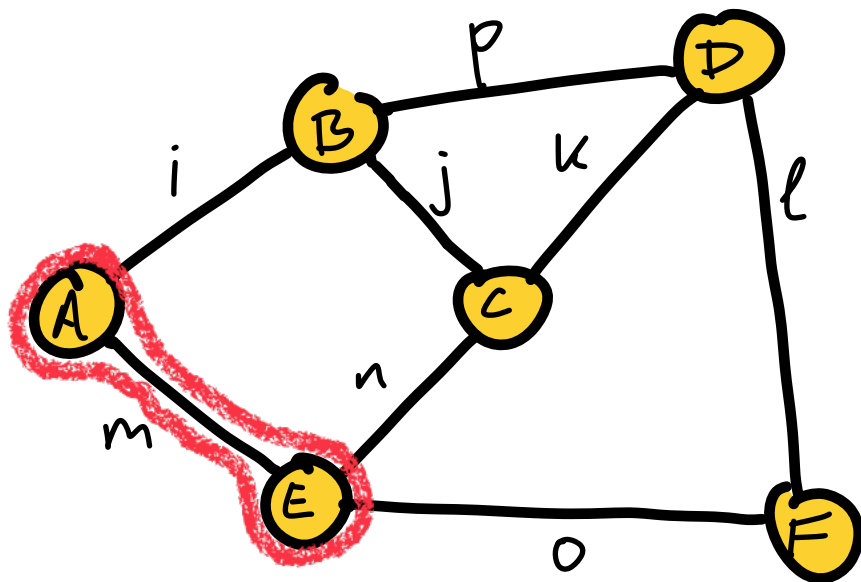
Singular Value Decomposition

$$\begin{aligned}
 & d \times d^{N-1} \\
 & \downarrow \\
 & d \times D_1 \quad A_1 S_1 V_1^\dagger \quad D_1 \times d^{N-1} \\
 & \swarrow \downarrow \\
 & D_1 d \times d^{N-2} \quad A_1 A_2 S_2 V_2^\dagger \\
 & \vdots \\
 & A_1 A_2 \dots A_{N-1} A_N
 \end{aligned}$$

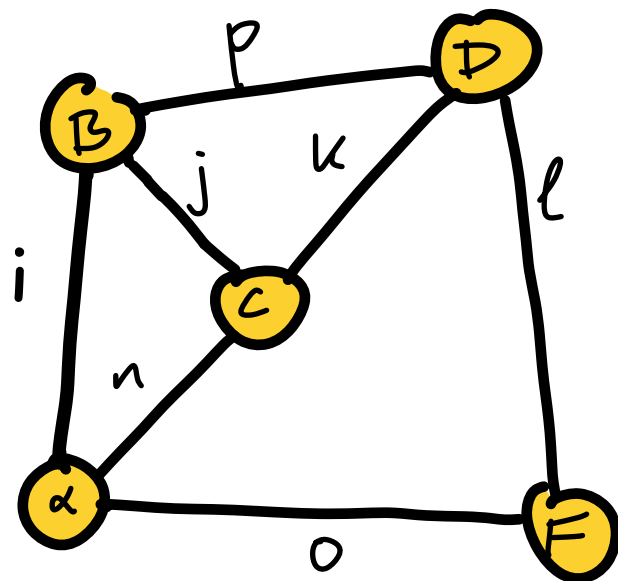




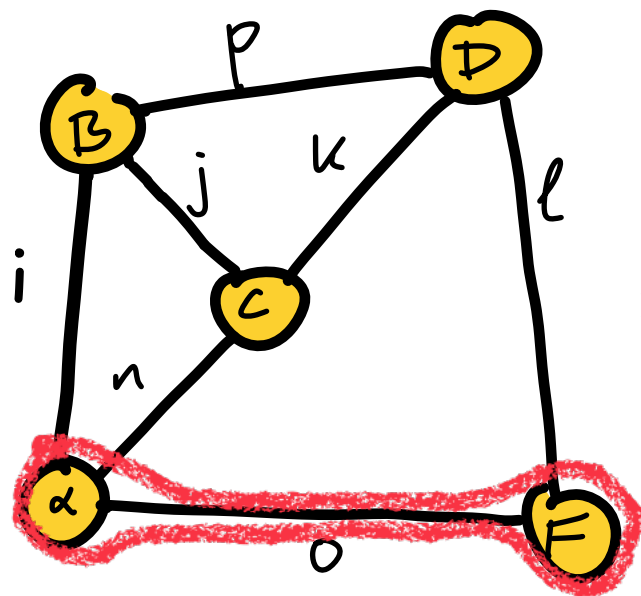
$$= \sum_{ijklmnop} A_{mi} B_{ijp} C_{jkn} D_{pkl} E_{mno} F_{ol}$$



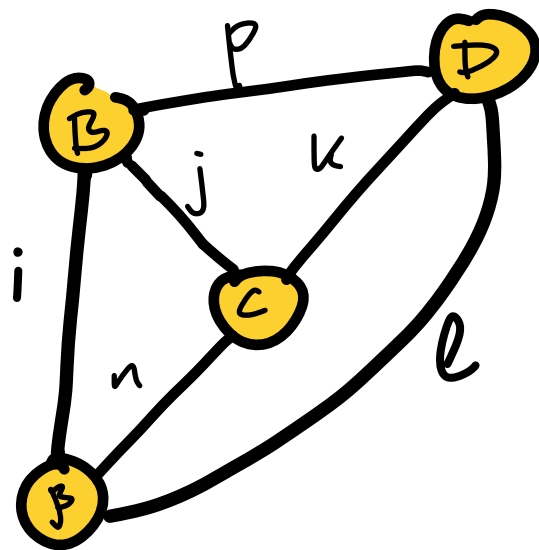
$$= \sum_{ijklmnop} A_{mi} B_{ijp} C_{jkn} D_{pkl} E_{mno} F_{ol}$$



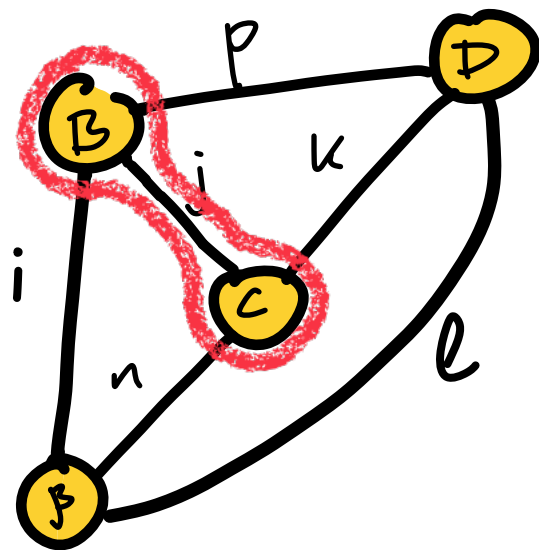
$$= \sum_{ijklno} \alpha_{ino} B_{ijp} C_{jkn} D_{pkl} F_{ol}$$



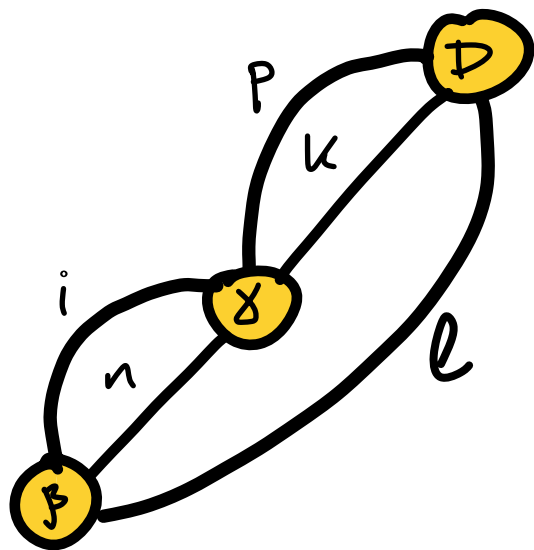
$$= \sum_{ijklno} \alpha_{ino} B_{ijp} C_{jkn} D_{pkl} F_{ol}$$



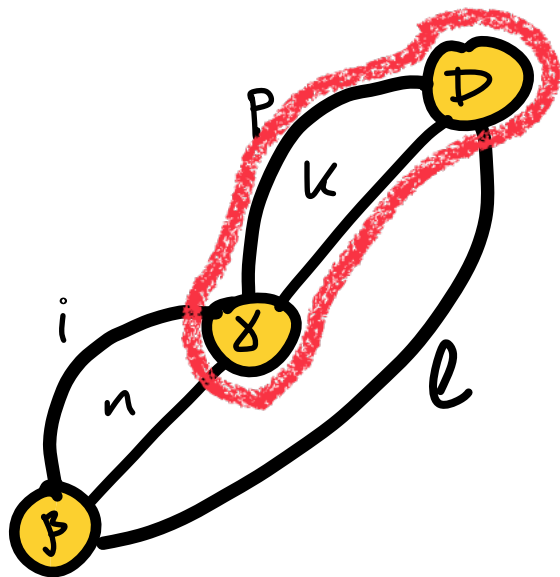
$$= \sum_{ijklnp} \beta_{inl} B_{ijp} C_{jkn} D_{pkl}$$



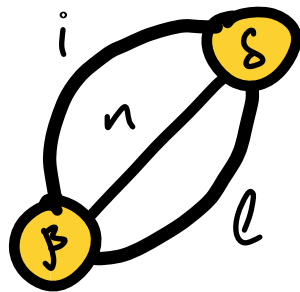
$$= \sum_{ijklnp} \beta_{inl} B_{ijp} C_{jkn} D_{pkl}$$



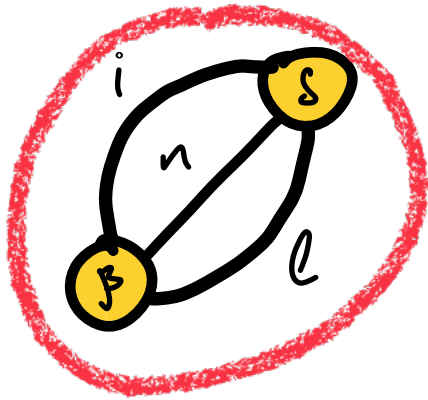
$$= \sum_{iklnp} \gamma_{inp} \beta_{inl} D_{pkl}$$



$$= \sum_{iklnp} \gamma_{inpk} \beta_{inl} D_{pkl}$$



$$= \sum_{i, l, n} \delta_{i, n, l} \beta_{i, n, l}$$

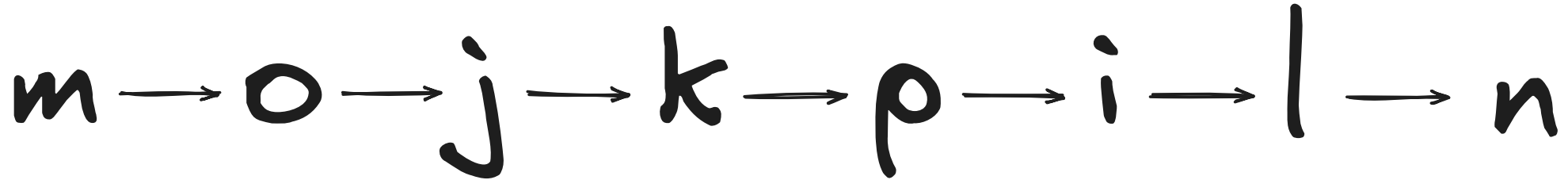


$$= \sum_{iln} \delta_{inl} \beta_{inl}$$

Scalar value

$$\text{ⓔ} = \mathcal{E}$$

Contraction path



- Find optimal contraction path for arbitrary TN $\Rightarrow$ NP-hard problem
 - Instead, **Graph-based heuristics and algorithms** (e.g., treewidth minimization, greedy search, dynamic programming, hypergraph partitioning) are used
- **Optimal contraction order dictates computational cost:**
 - sequence in which tensors are contracted determines the scaling
 - a poor path can turn a polynomial problem into an exponential one.
- **For 1D TN structures:** The contraction is *naturally efficient*—a simple left-to-right or right-to-left sweep gives linear-time contraction with minimal intermediate growth.

Tensor Networks for ML

Machine Learning

computers learn to perform tasks without being explicitly programmed

Ingredients of learning problem

DATA

$$\mathcal{D} = \{(x^1, y^1), \dots, (x^M, y^M)\}$$

$$\mathcal{D} = \{x^1, \dots, x^M\}$$

MODEL

deterministic / probabilistic

OPTIMIZATION

- objective function
- risk minimization

Types of models

Supervised Learning

$$f : X \rightarrow Y$$

deterministic

$$p_{\theta} : X \times Y \rightarrow [0,1], \quad p_{\theta}(x, y)$$

$$p_{\theta} : X \times Y \rightarrow [0,1], \quad p_{\theta}(y | x)$$

Unsupervised Learning

$$p_{\theta} : X \rightarrow [0,1], \quad p_{\theta}(x)$$

probabilistic

Tensor Networks

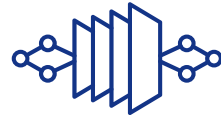
Classification



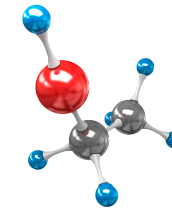
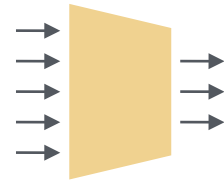
Anomaly
detection



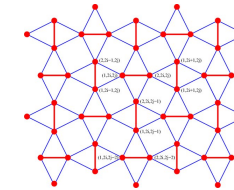
Generative
modeling



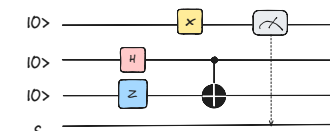
Data reduction
Model compression



Quantum
chemistry



Quantum
systems



Benchmark
quantum
algorithms



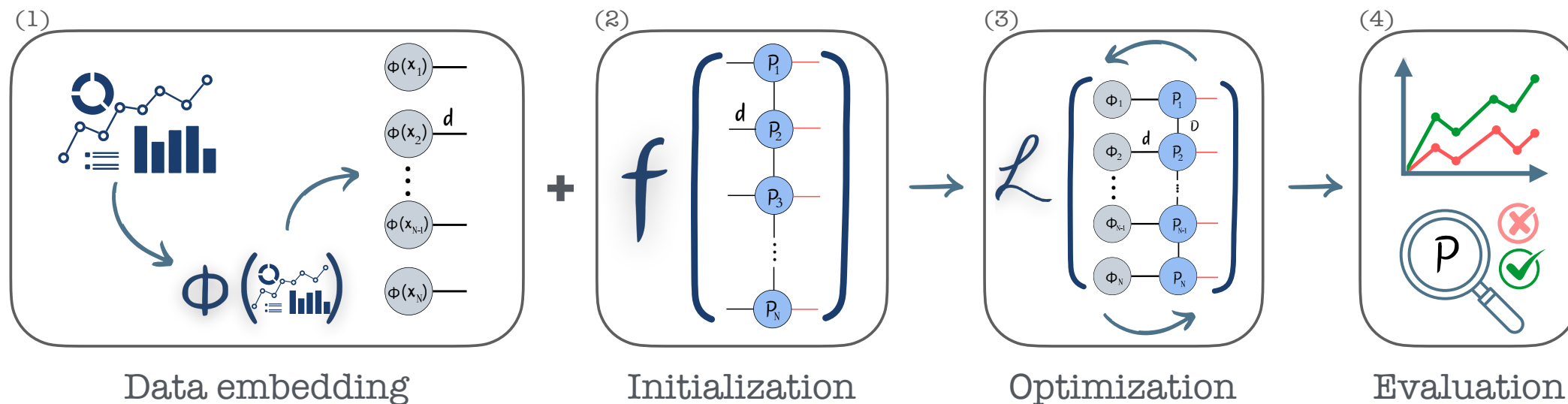
planqc

classical ML

quantum ML

Tensor Networks as AI models

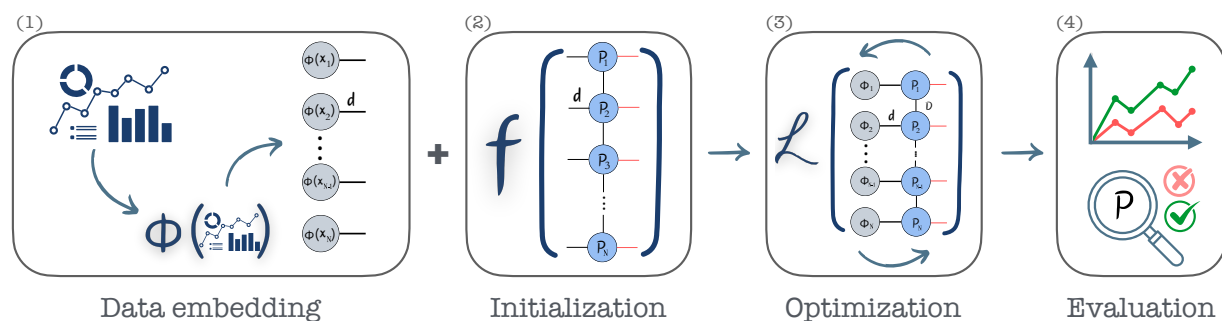
1) Parametrized TN for ML task



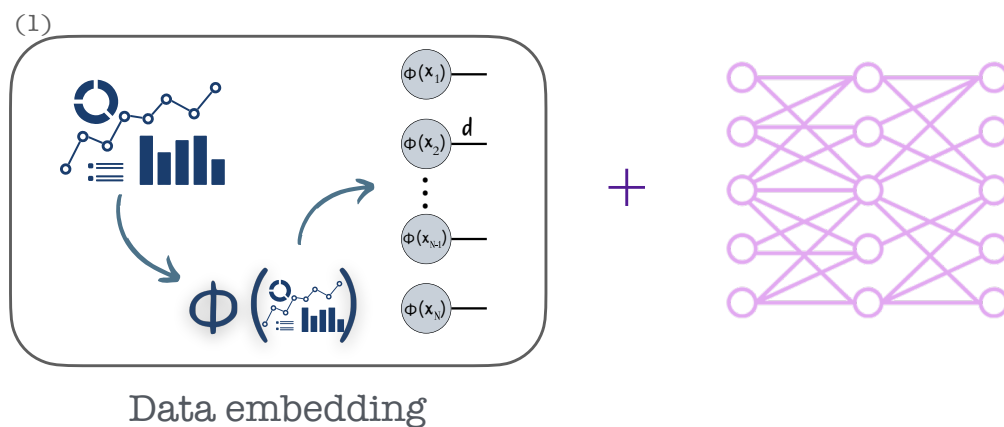
tn4ml: Tensor Network Training and Customization for Machine Learning [2502.13090]

Tensor Networks as AI models

1) Parametrized TN for ML task

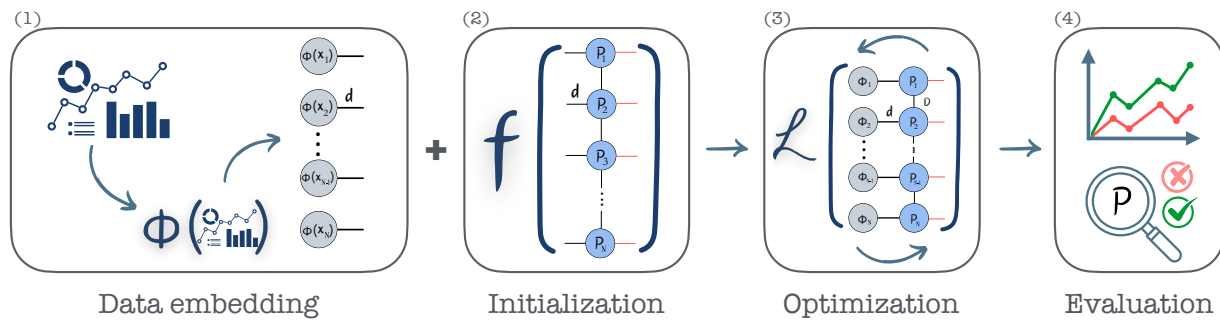


2) Data compression with TN + NN

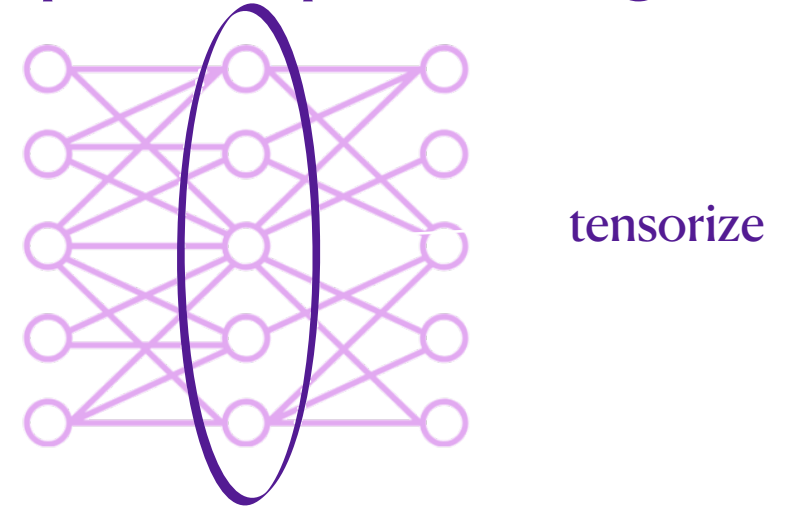


Tensor Networks as AI models

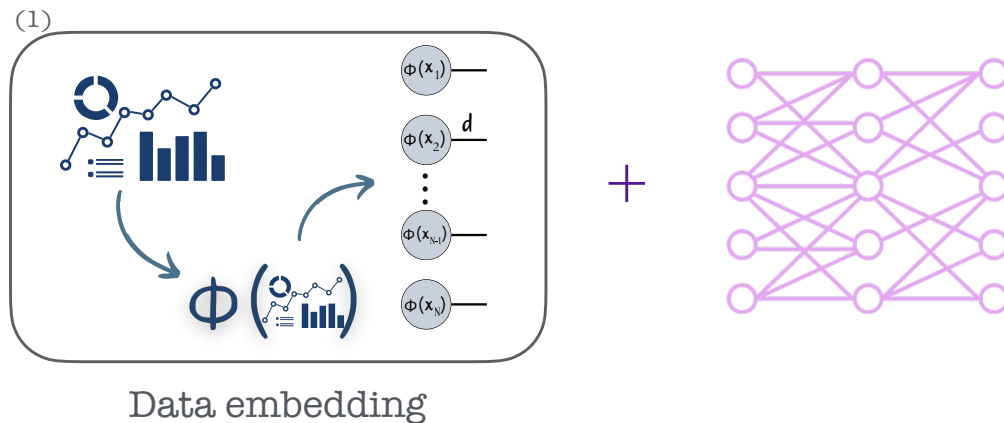
1) Parametrized TN for ML task



3) Compress NN post-training

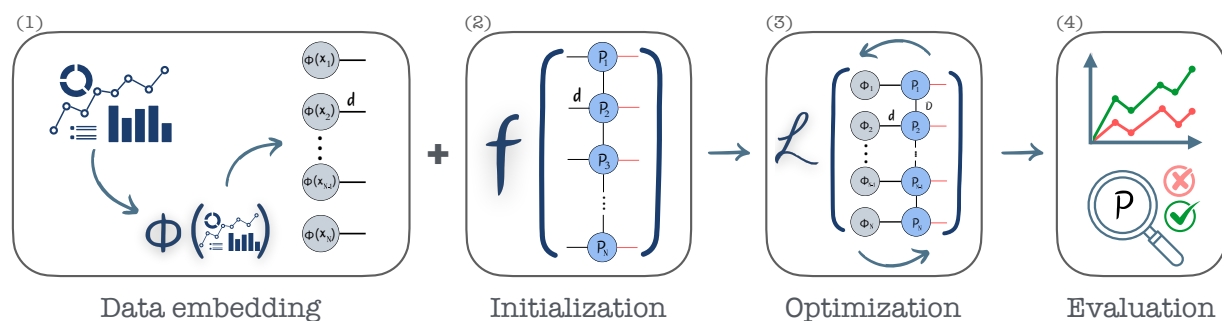


2) Data compression with TN + NN

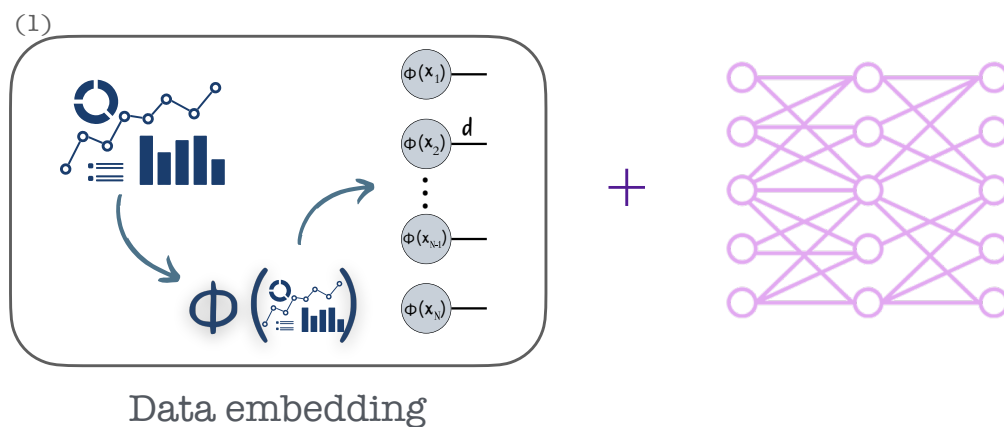


Tensor Networks as AI models

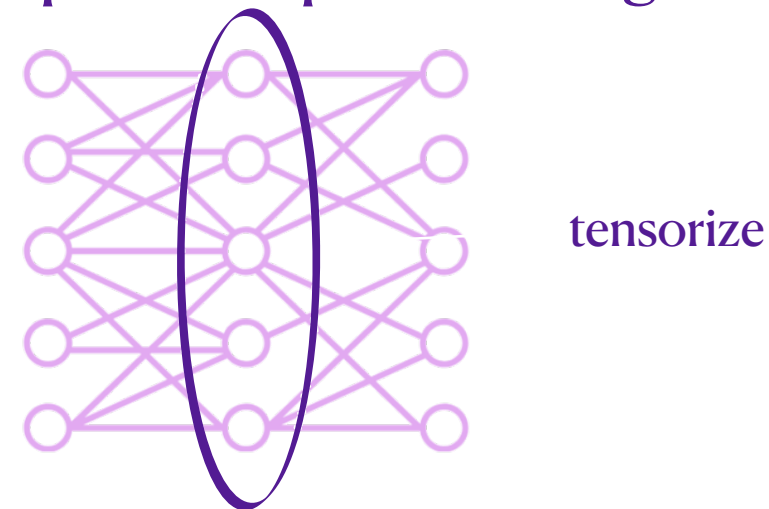
1) Parametrized TN for ML task



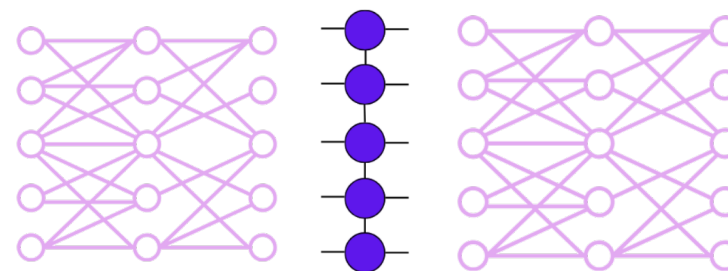
2) Data compression with TN + NN



3) Compress NN post-training

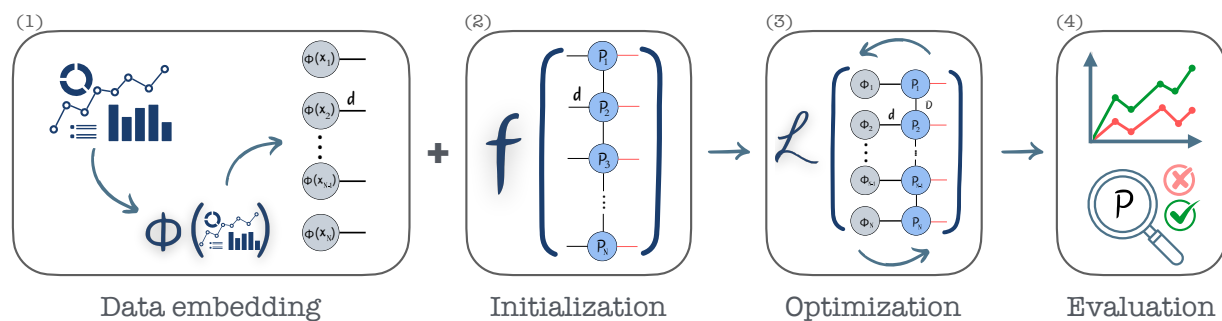


4) Hybrid TN-NN structures

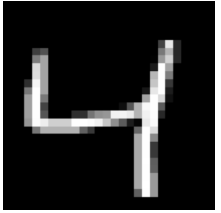


Tensor Networks as AI models

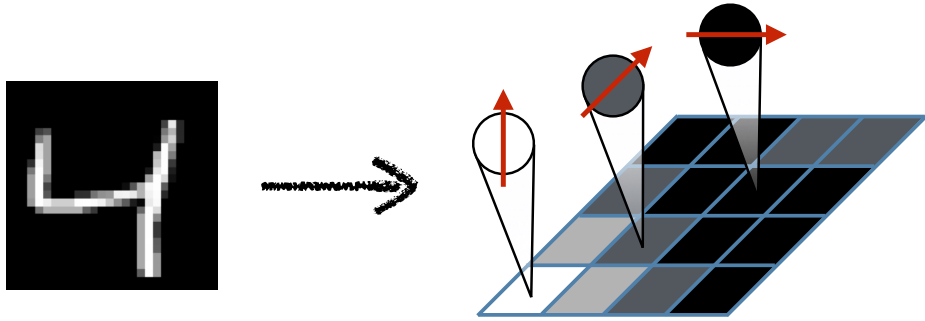
1) Parametrized TN for ML task



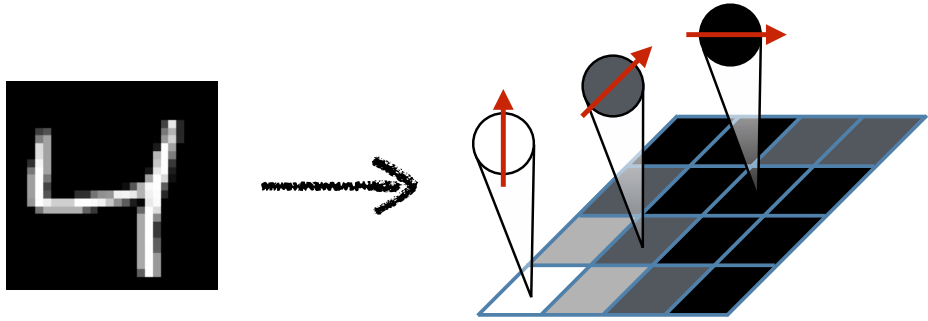
MNIST clasiffication



MNIST clasiffication



MNIST clasiffication

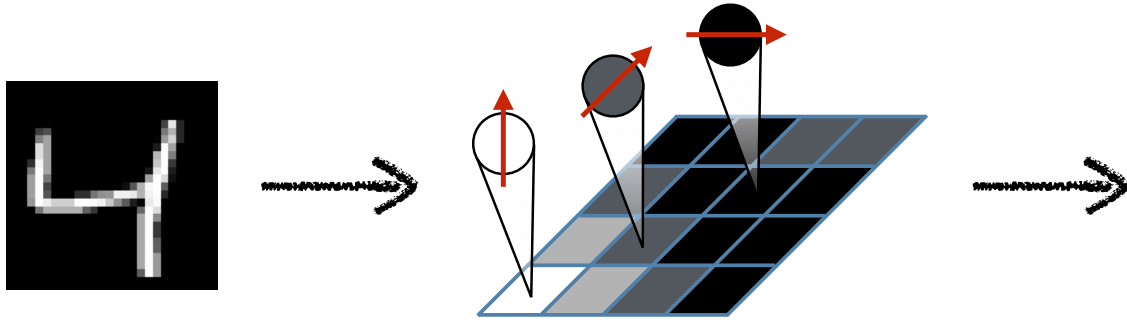


$$\phi(x_i) = \left[\cos\left(\frac{\pi}{2}x_i\right), \sin\left(\frac{\pi}{2}x_i\right) \right]$$

Feature function

(Defines physical dimension)

MNIST clasiffication

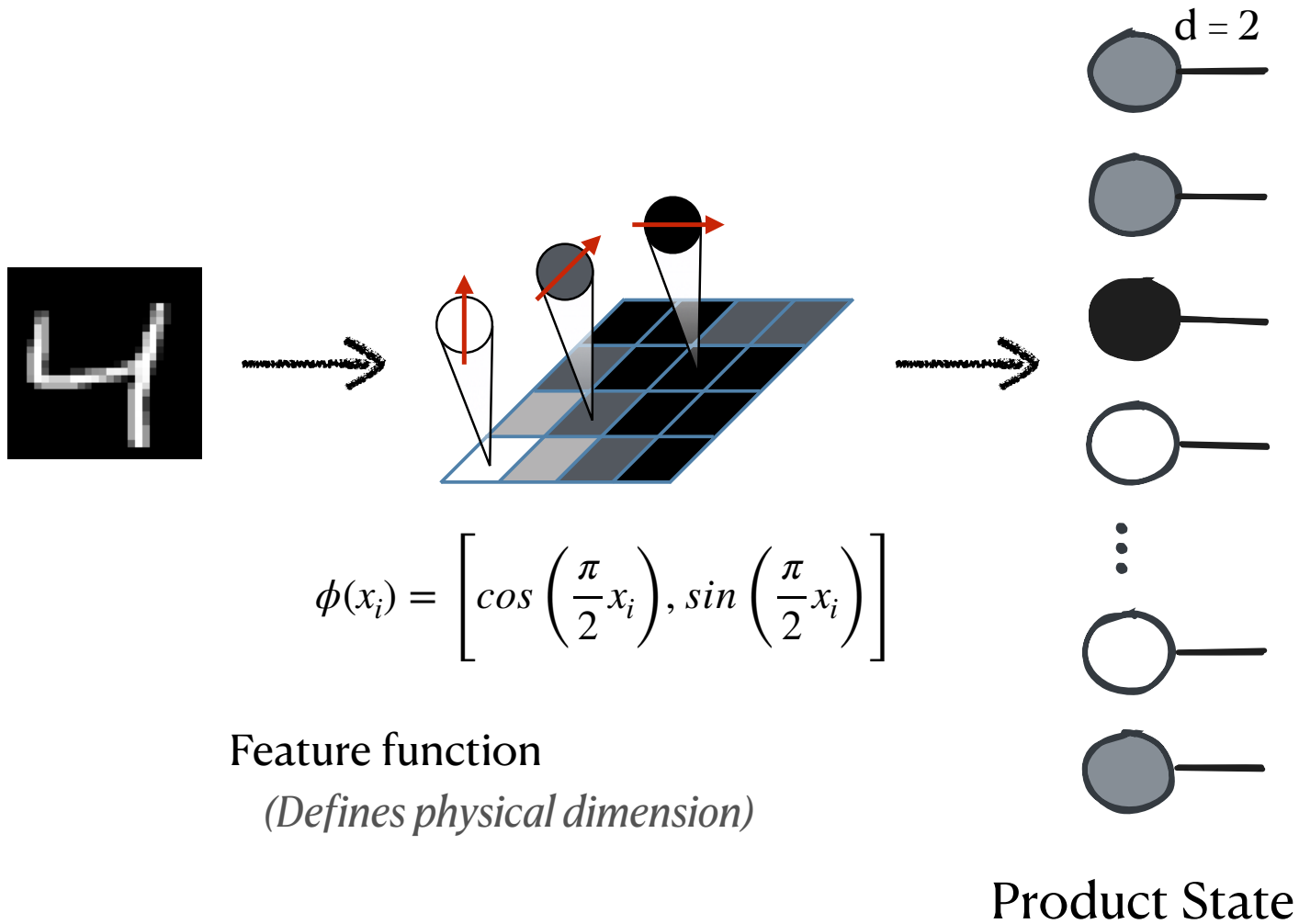


$$\phi(x_i) = \left[\cos\left(\frac{\pi}{2}x_i\right), \sin\left(\frac{\pi}{2}x_i\right) \right]$$

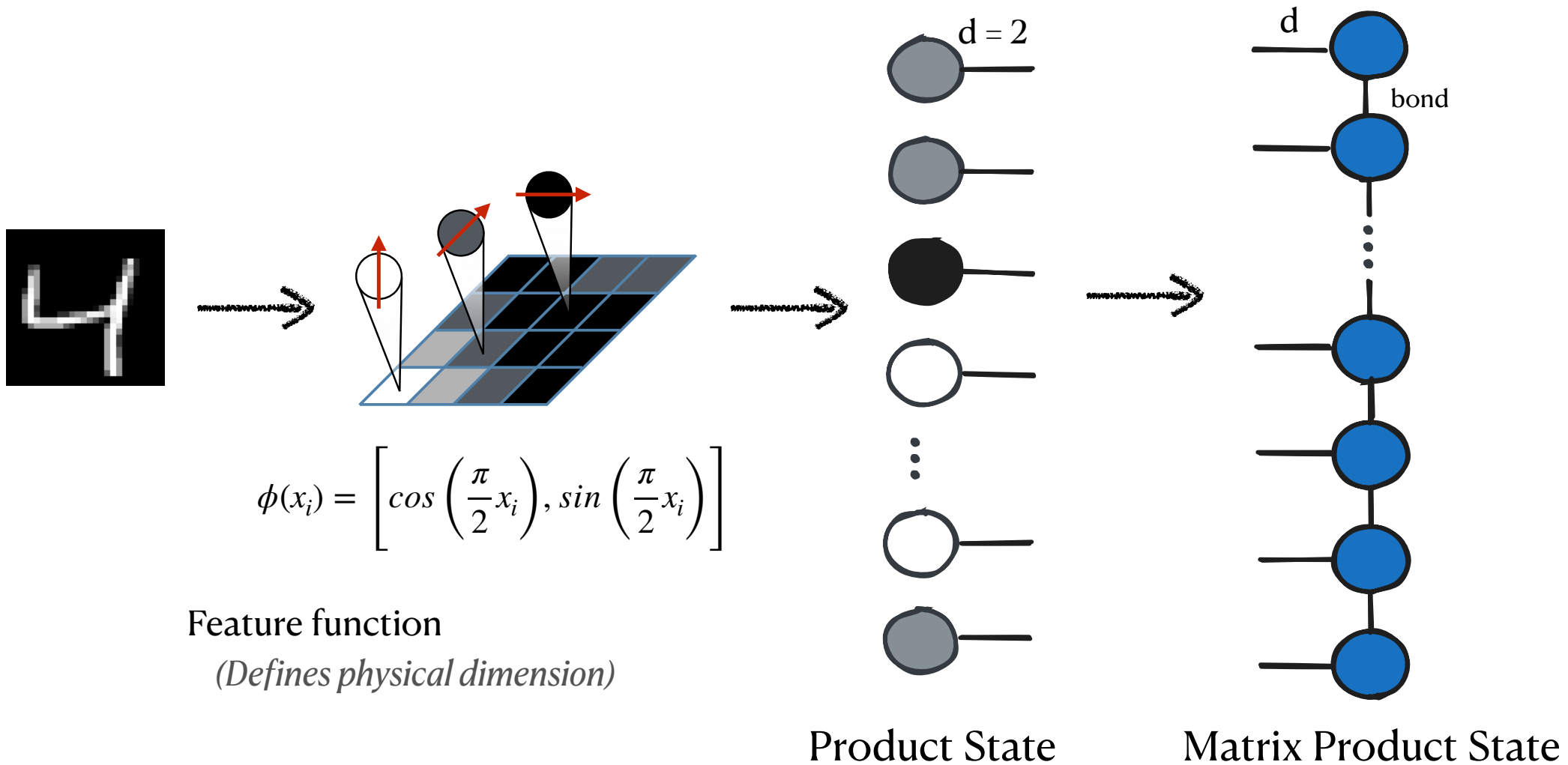
Feature function

(Defines physical dimension)

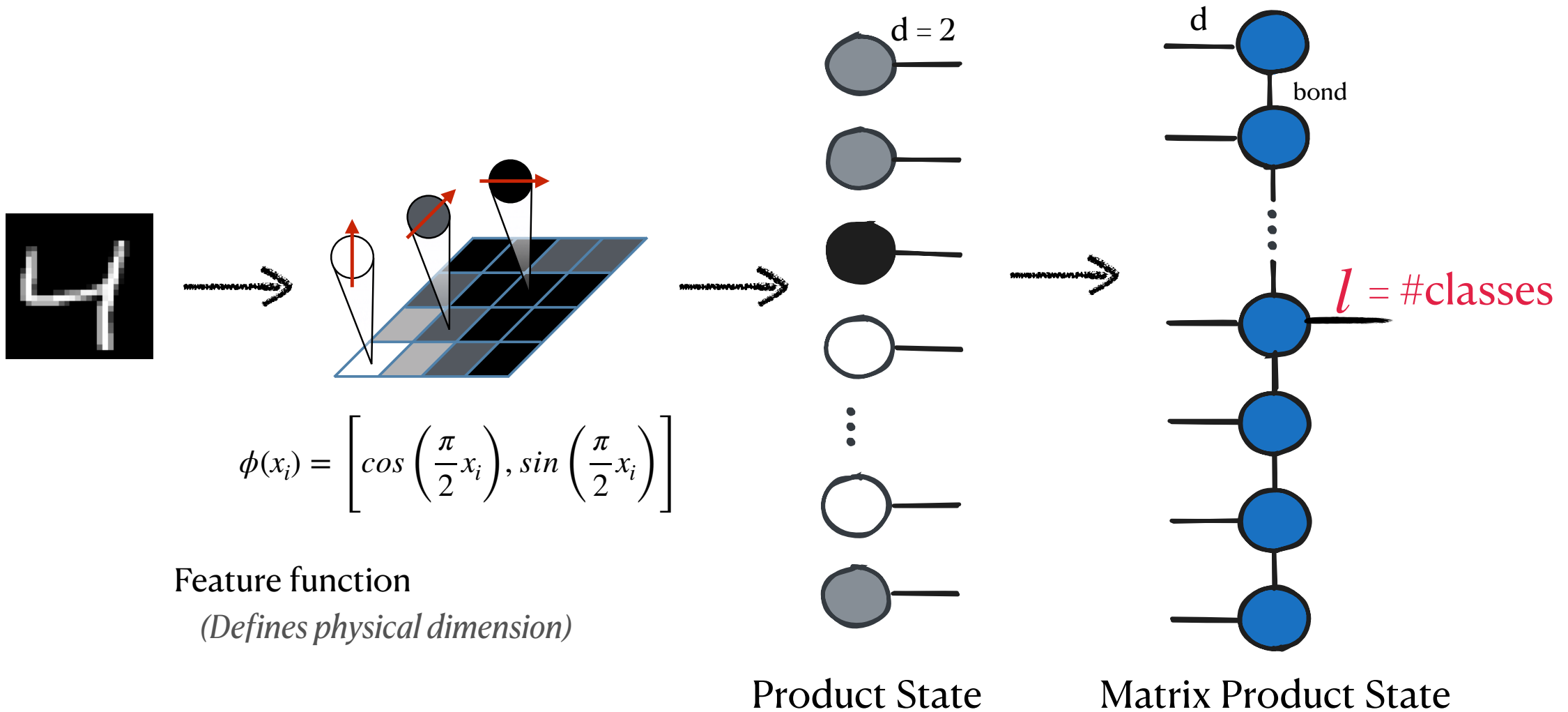
MNIST clasiffication



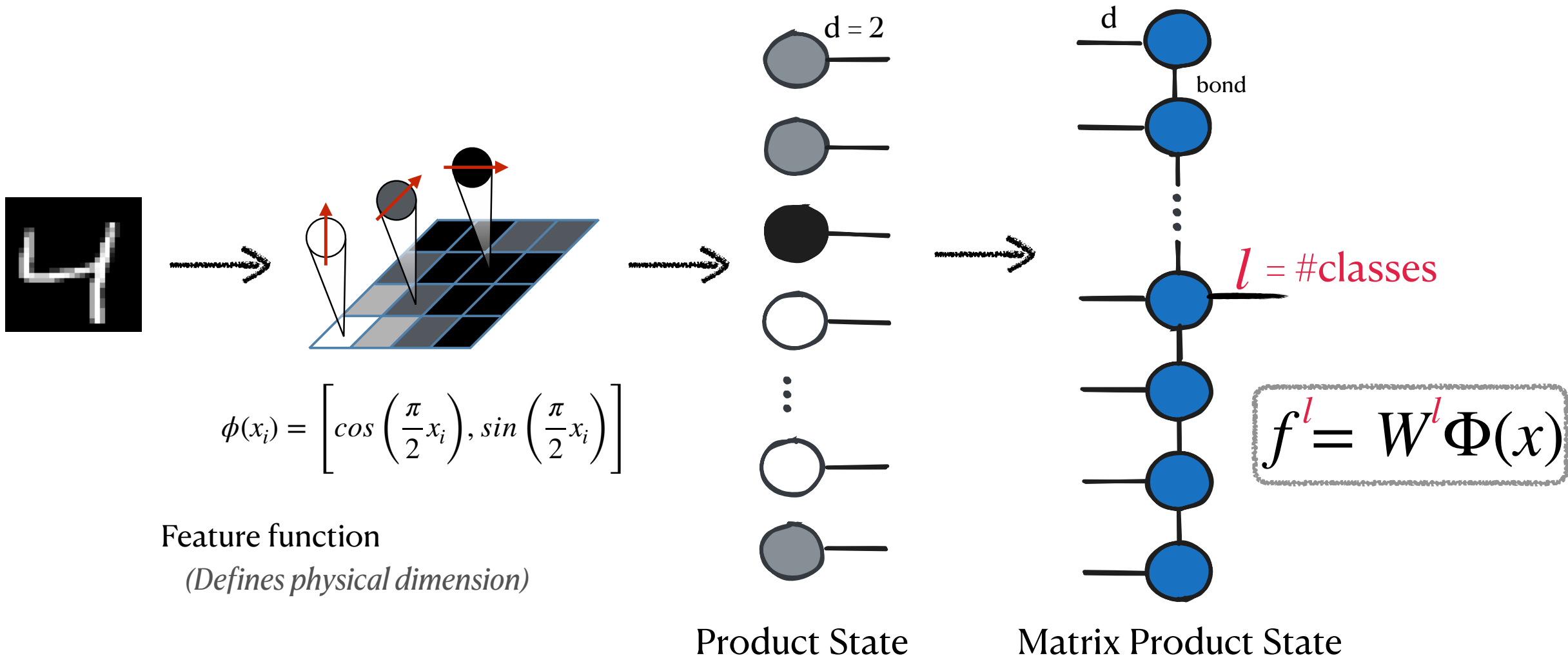
MNIST clasiffication



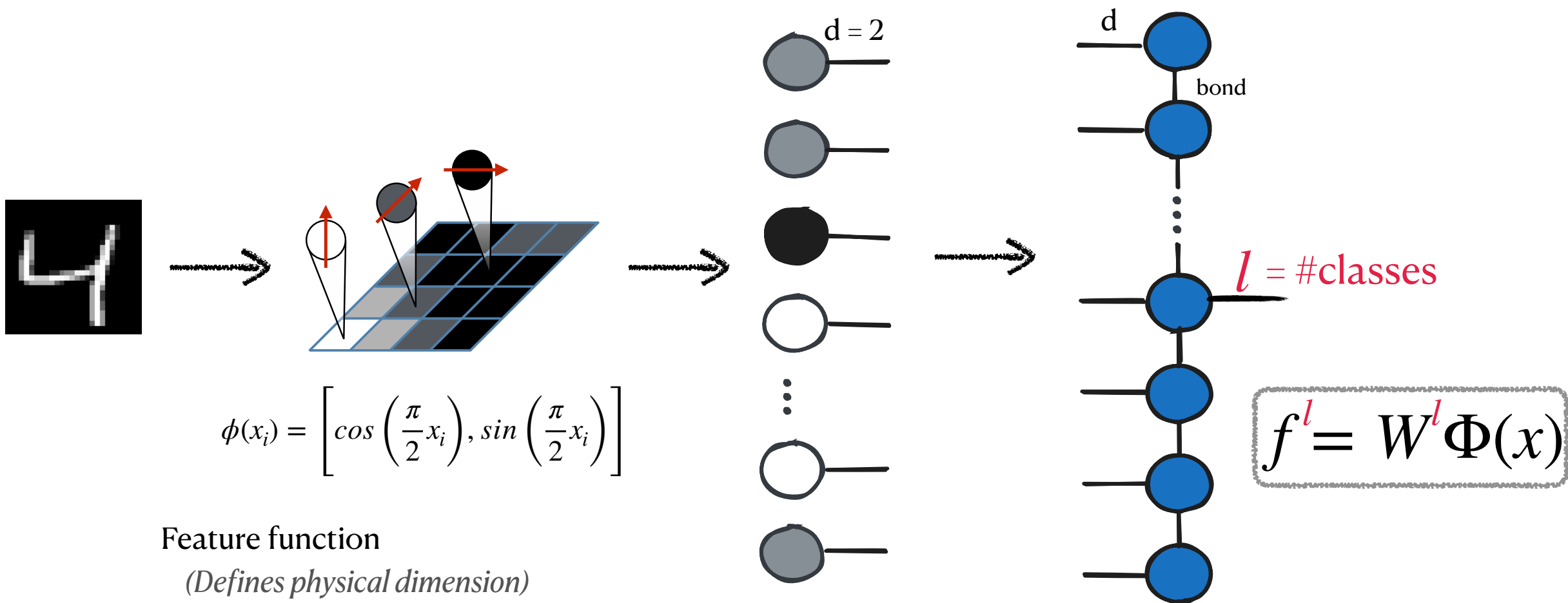
MNIST classification



MNIST clasiffication



MNIST clasiffication

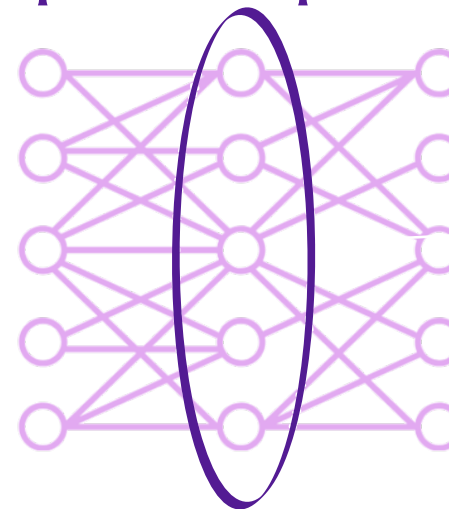


Supervised Learning with Quantum-Inspired Tensor Networks [1605.05775]

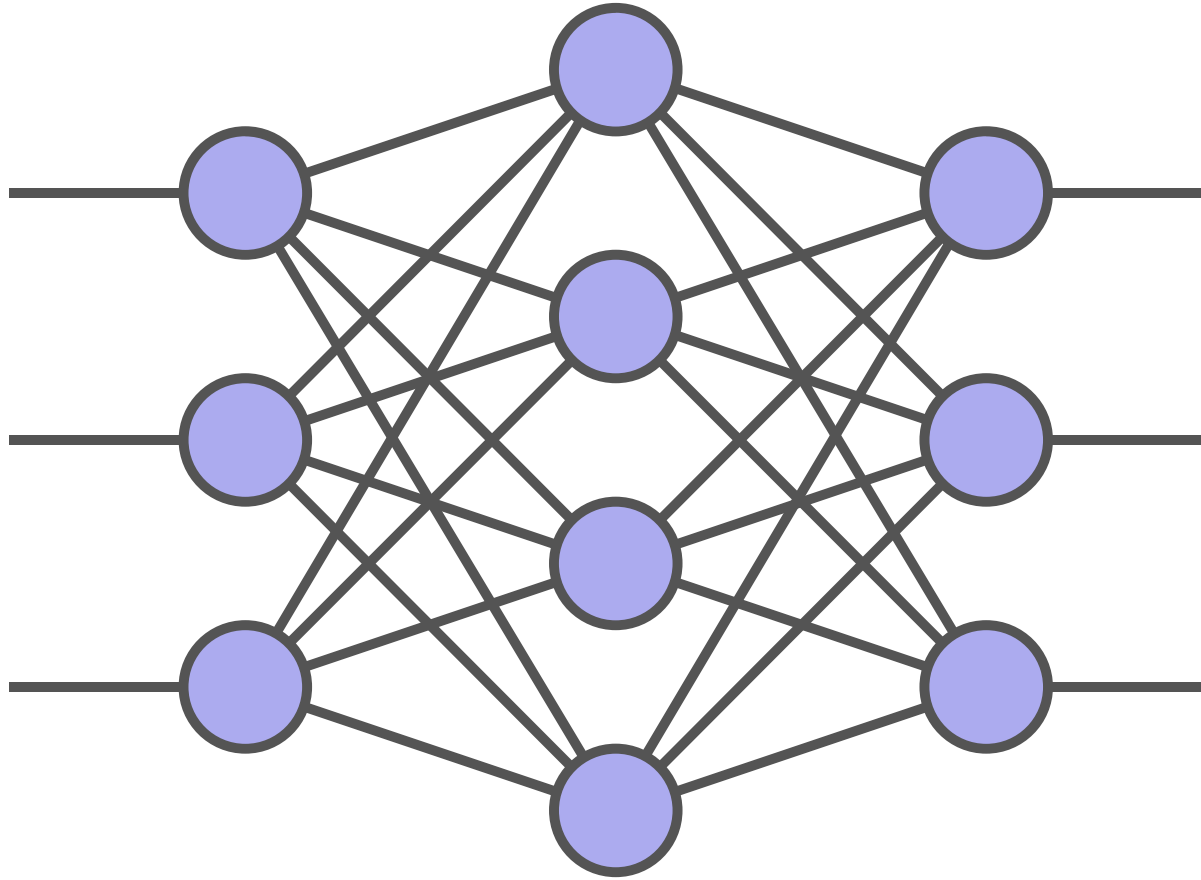
tn4ml: Tensor Network Training and Customization for Machine Learning [2502.13090]

Tensor Networks as AI models

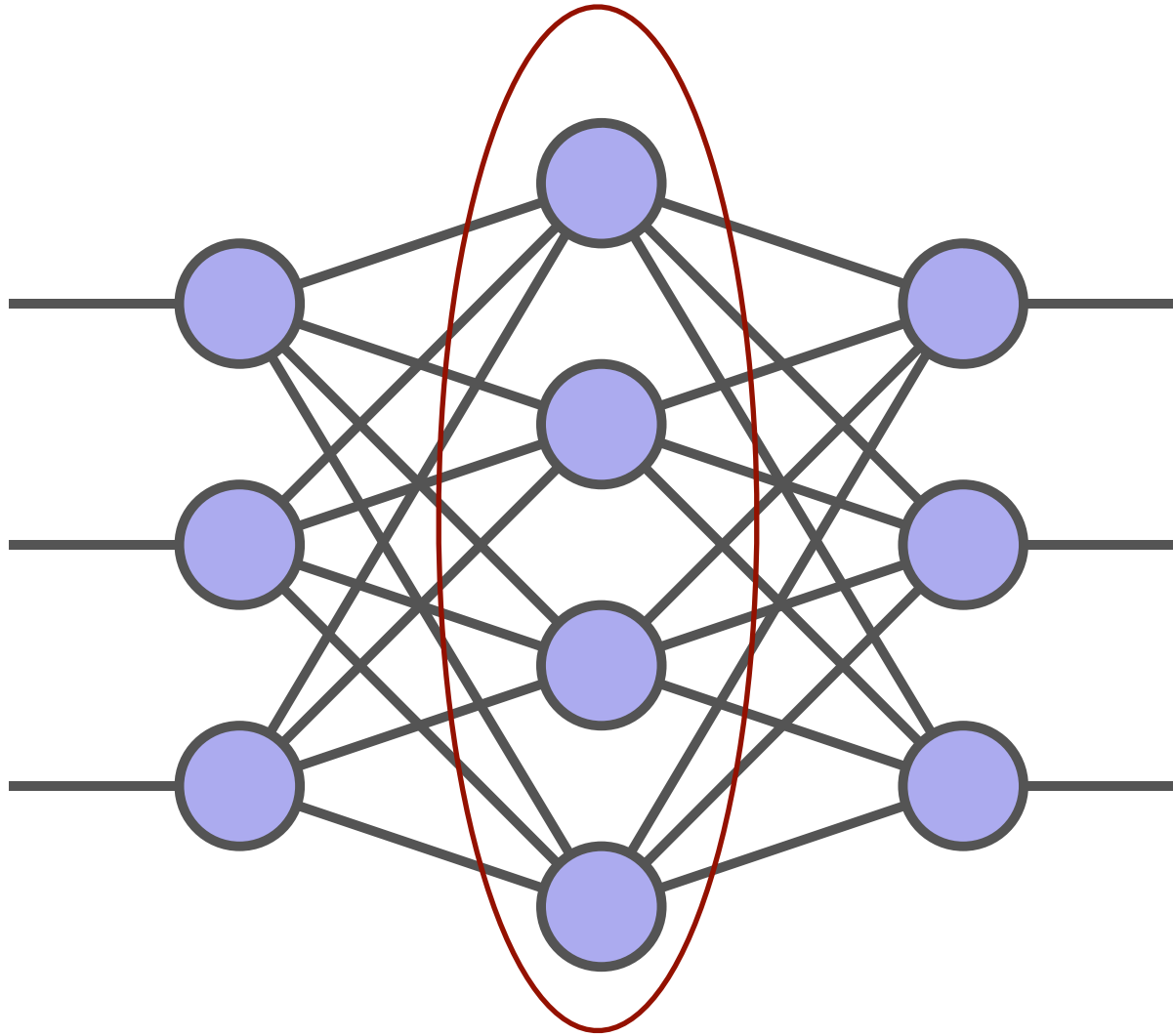
3) Compress NN post-training



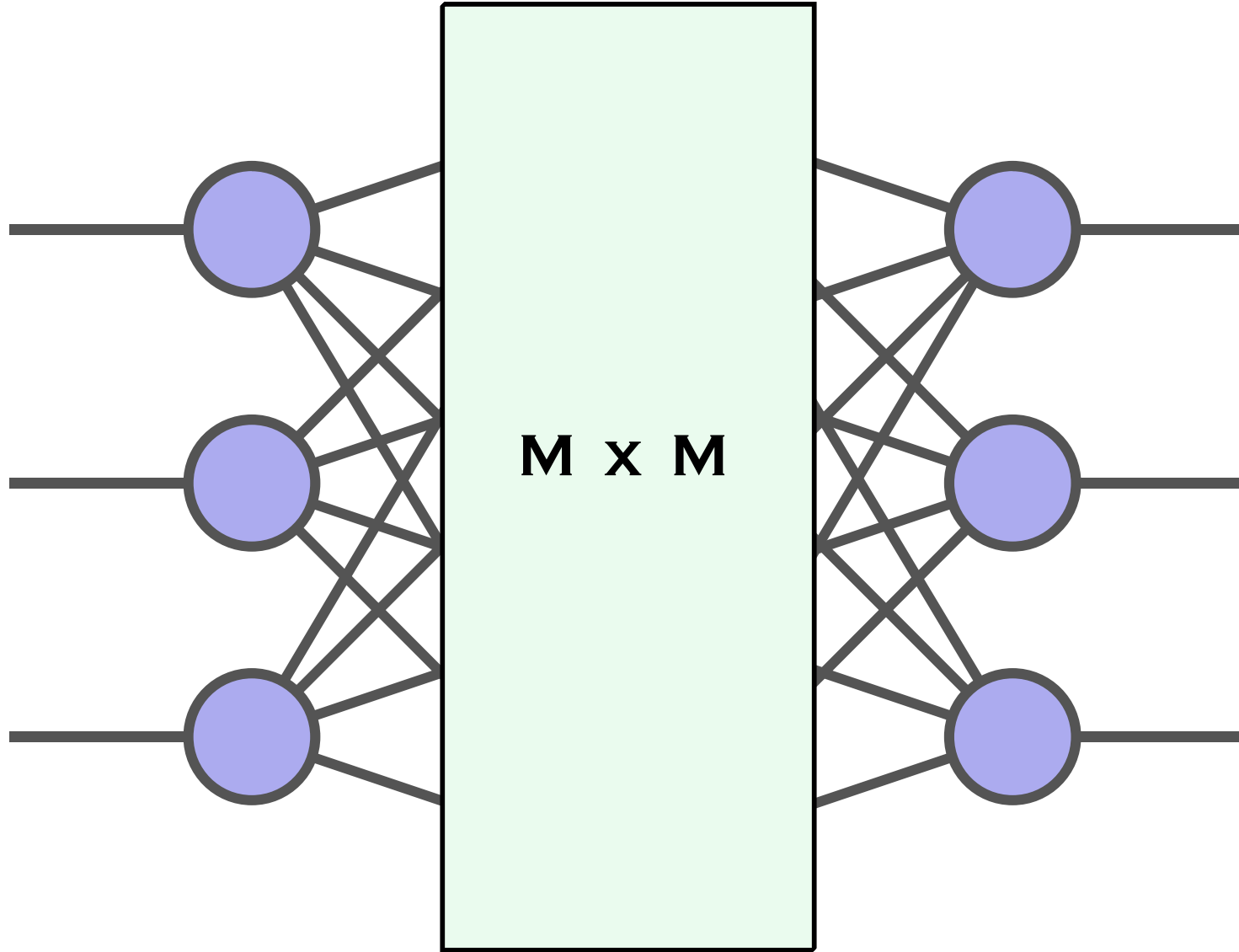
tensorize

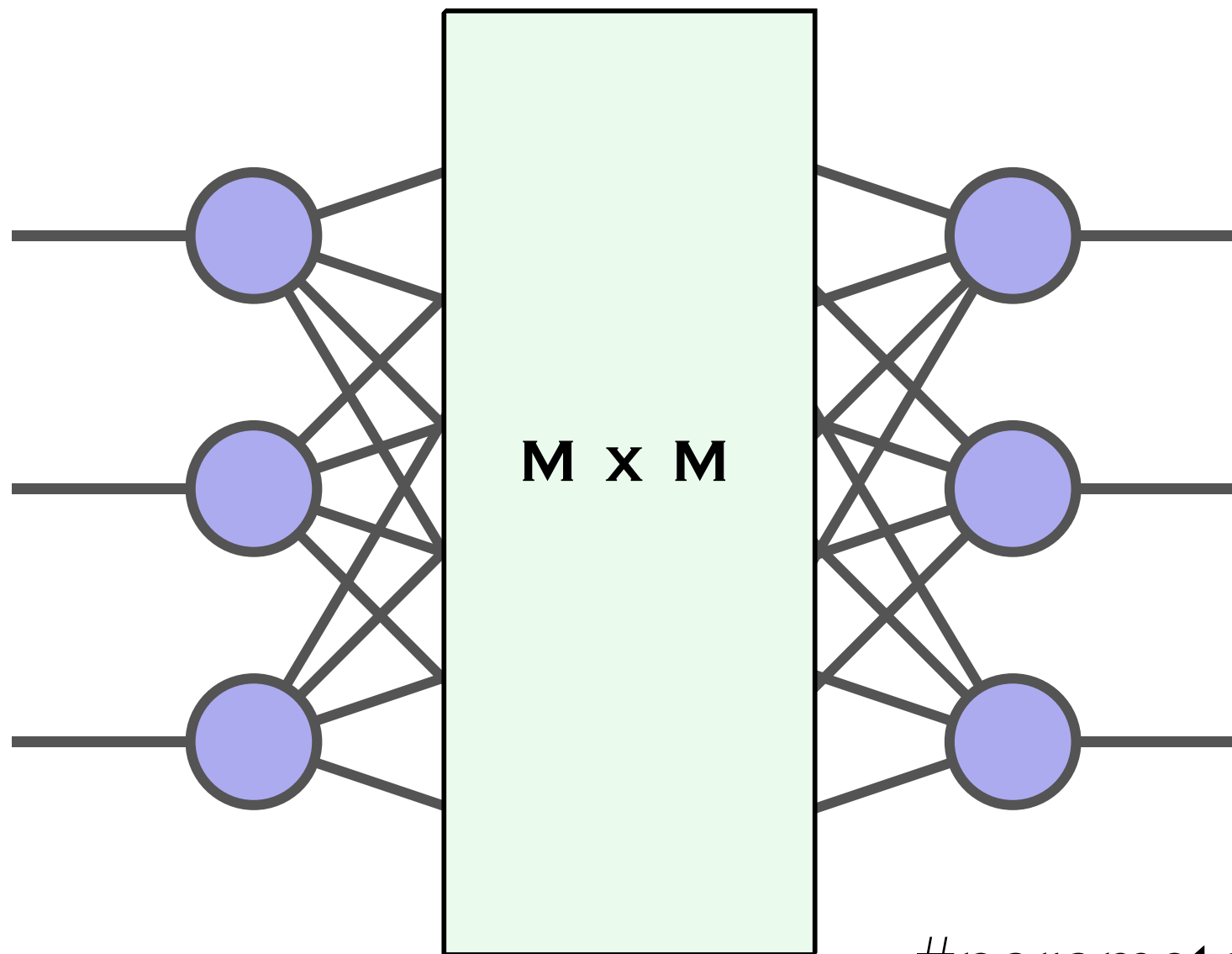


$$y = Wx + b$$

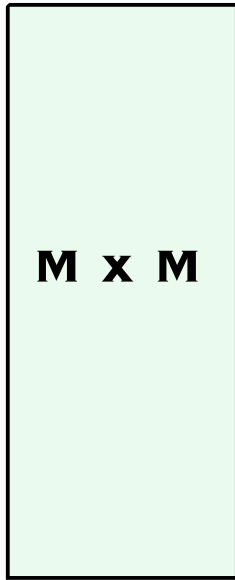


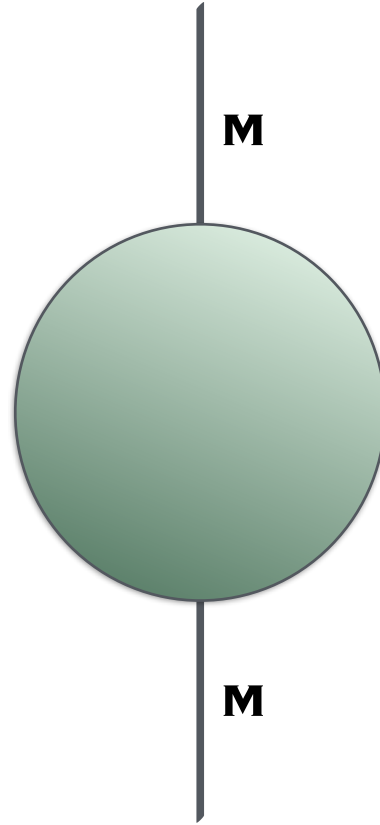
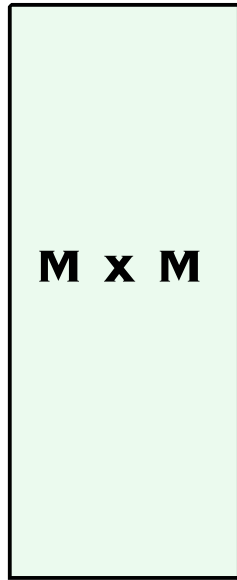
$$y = \mathbf{W}x + b$$

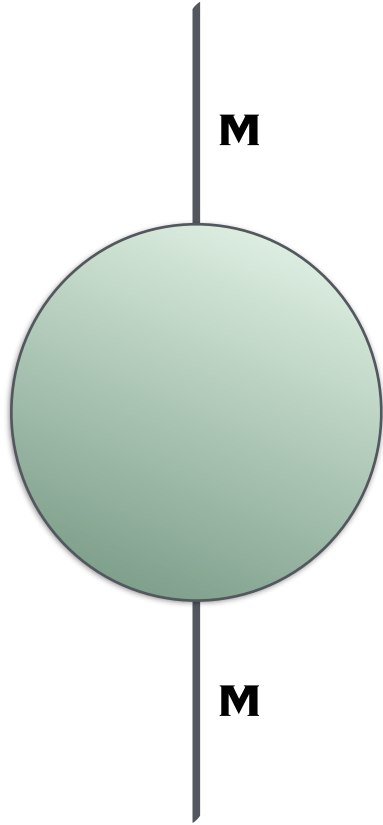


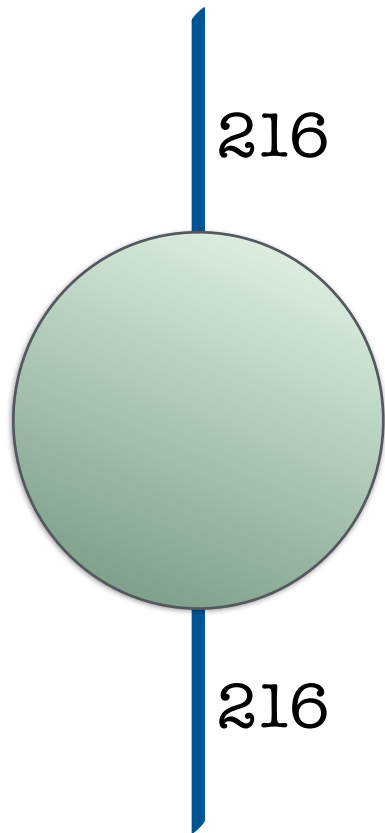


$$\# \text{parameters} = M^2$$

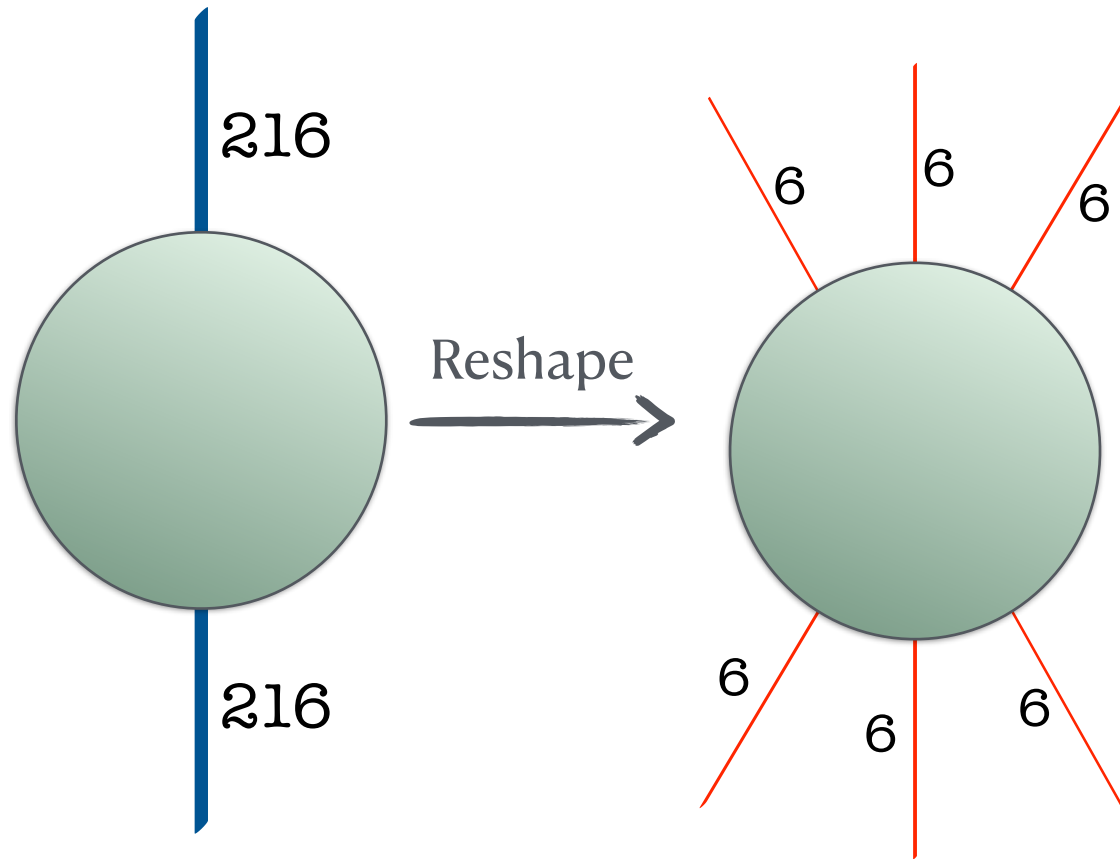






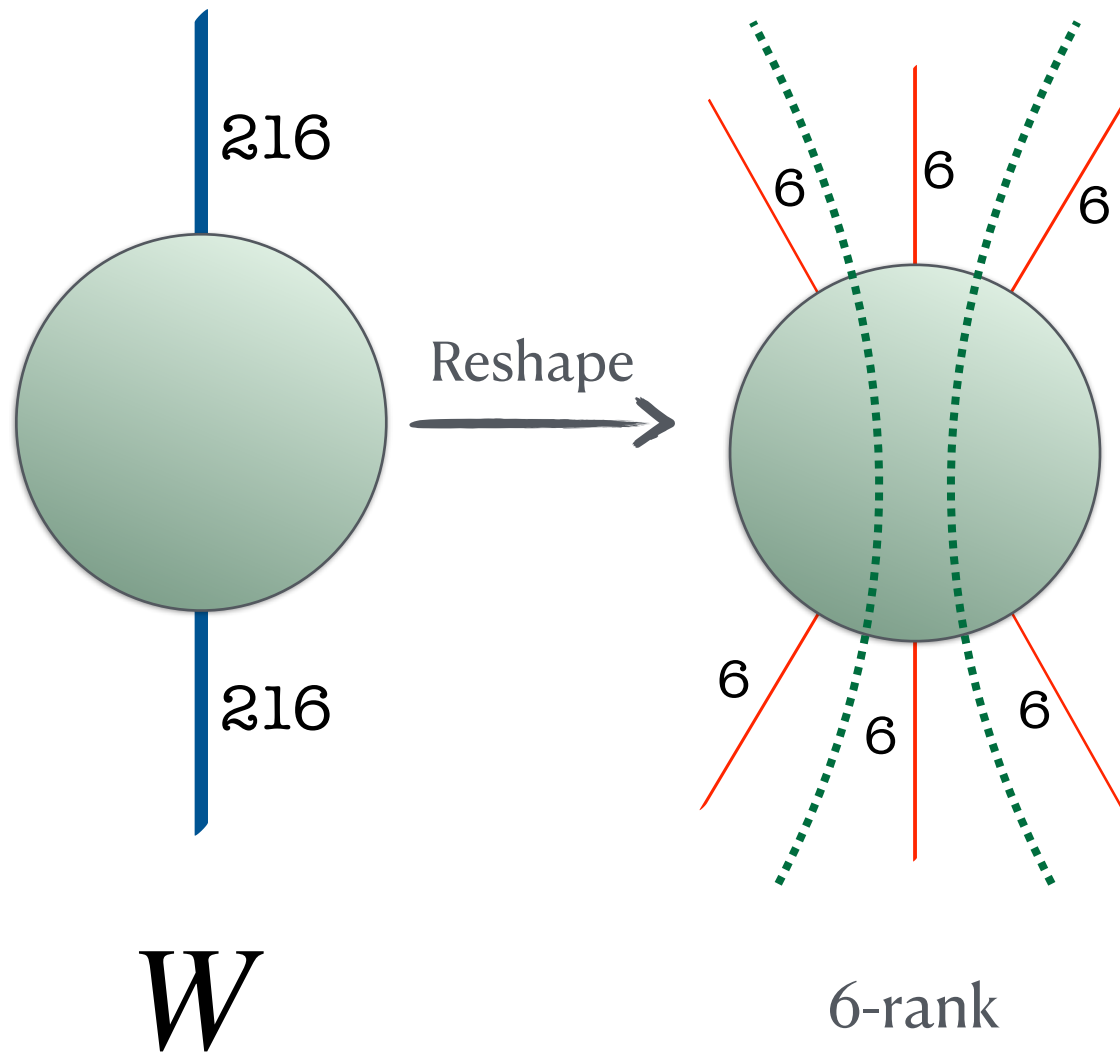


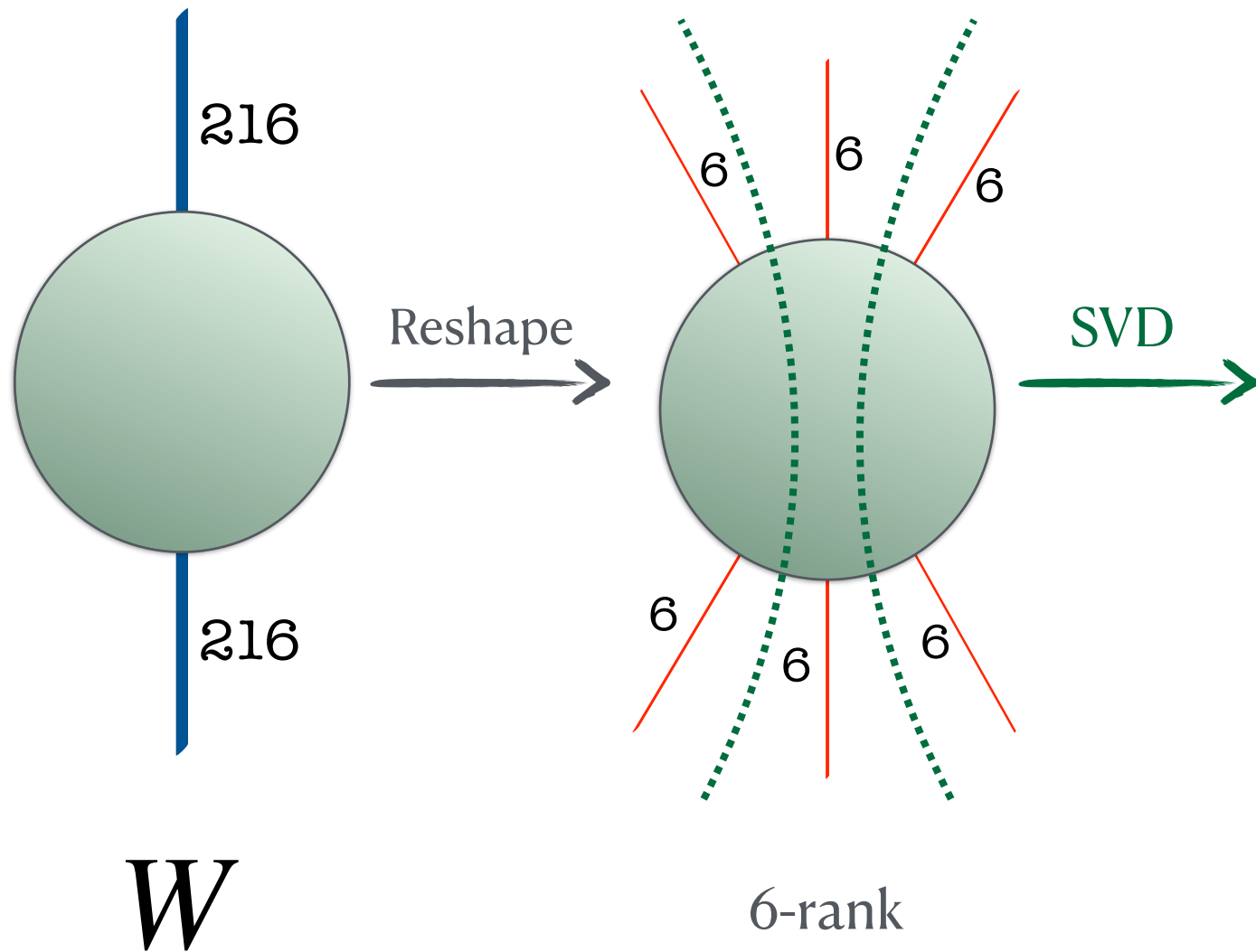
W

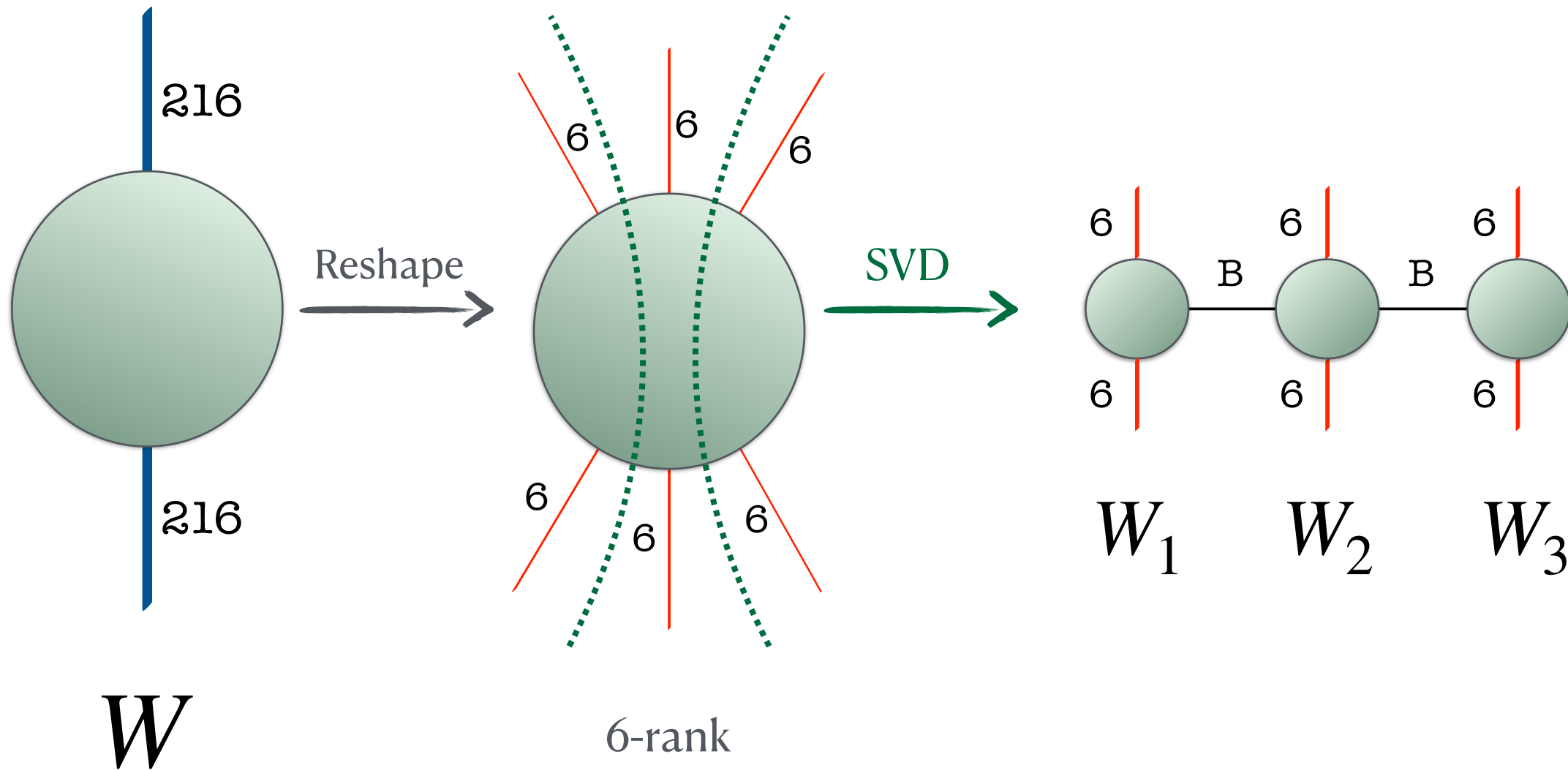


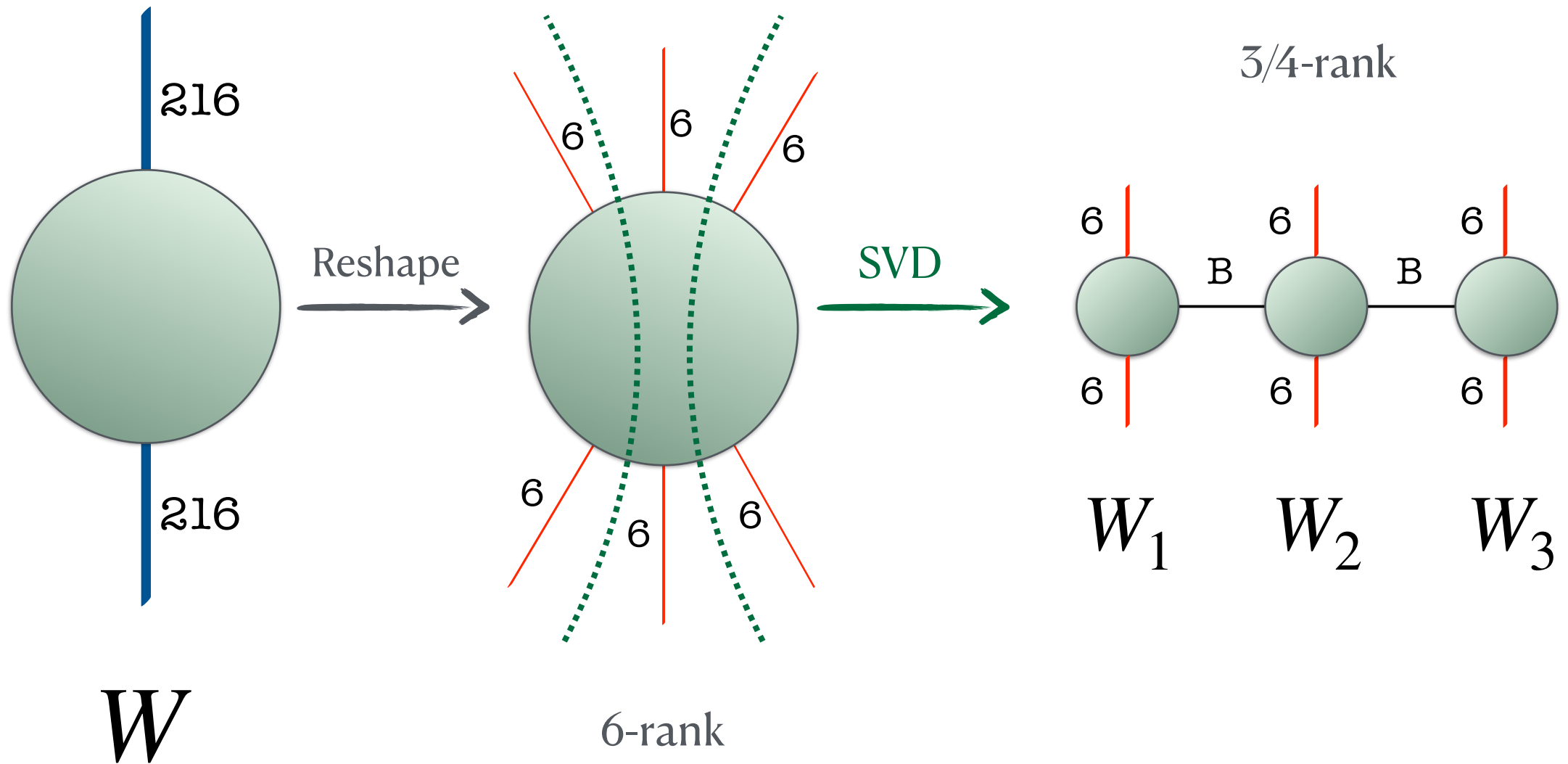
W

6-rank

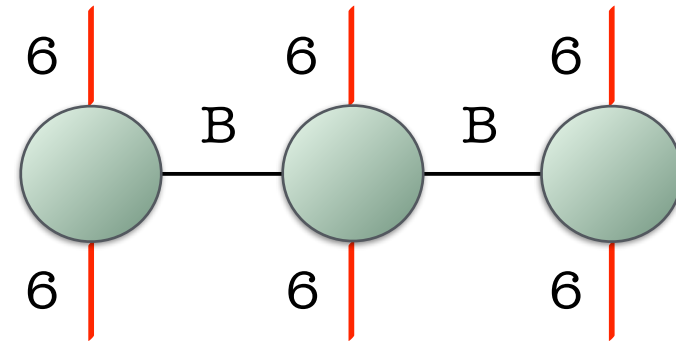
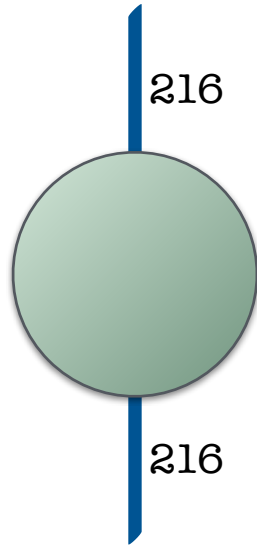




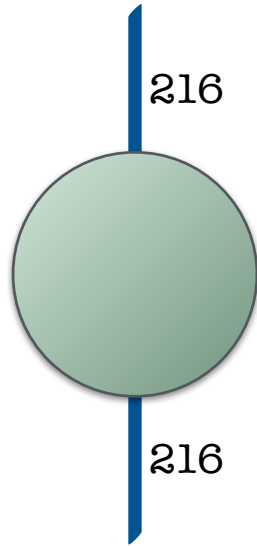




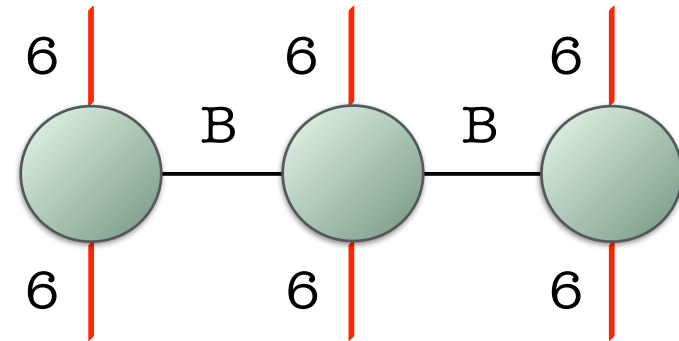
#params



#params

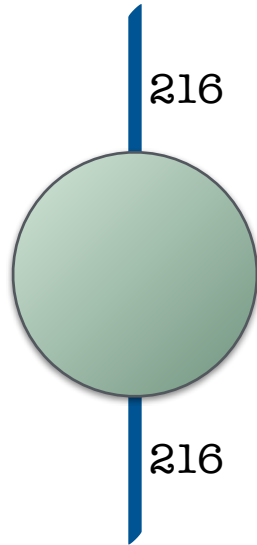


$$216^2$$

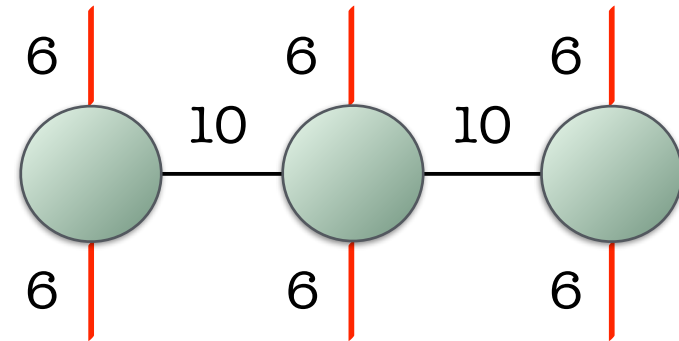


$$3 \times 6^2 \times B^2$$

#params

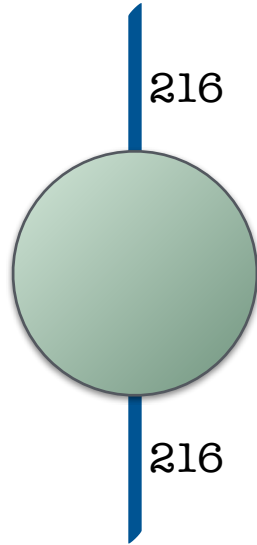


$$216^2$$

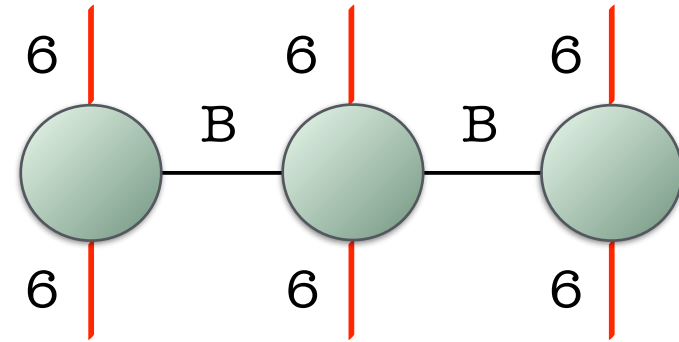


$$3 \times 6^2 \times 10^2$$

#params

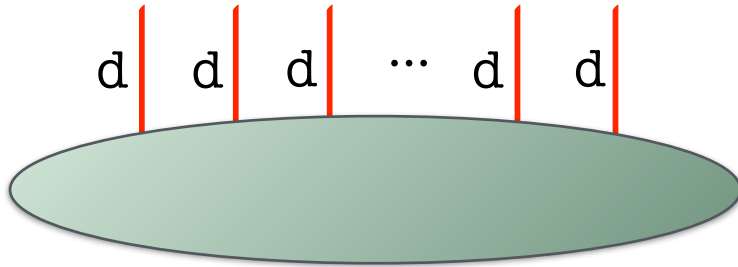


46656

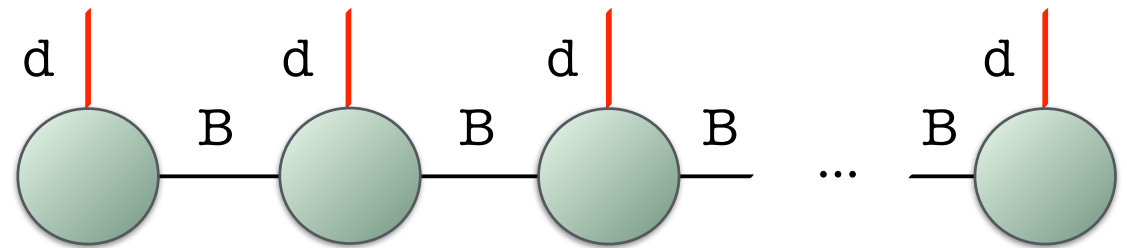


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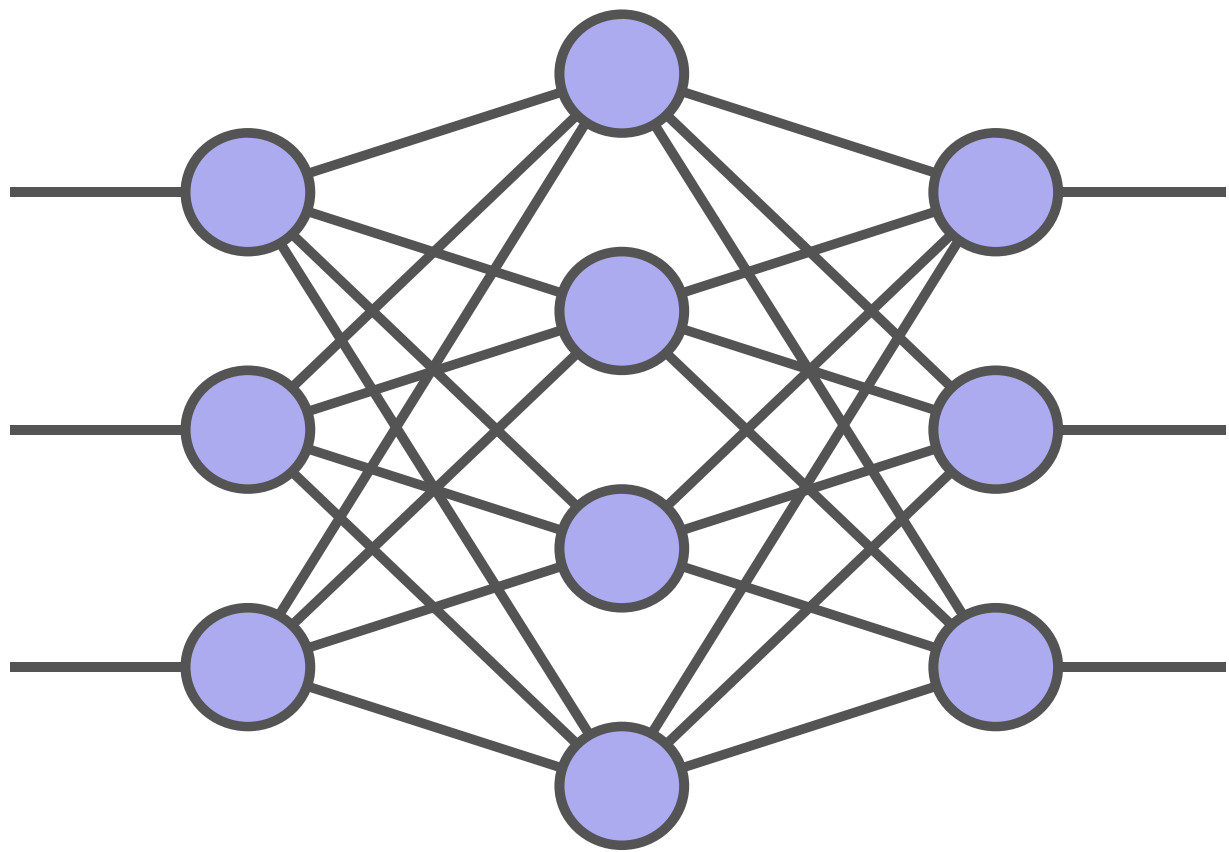
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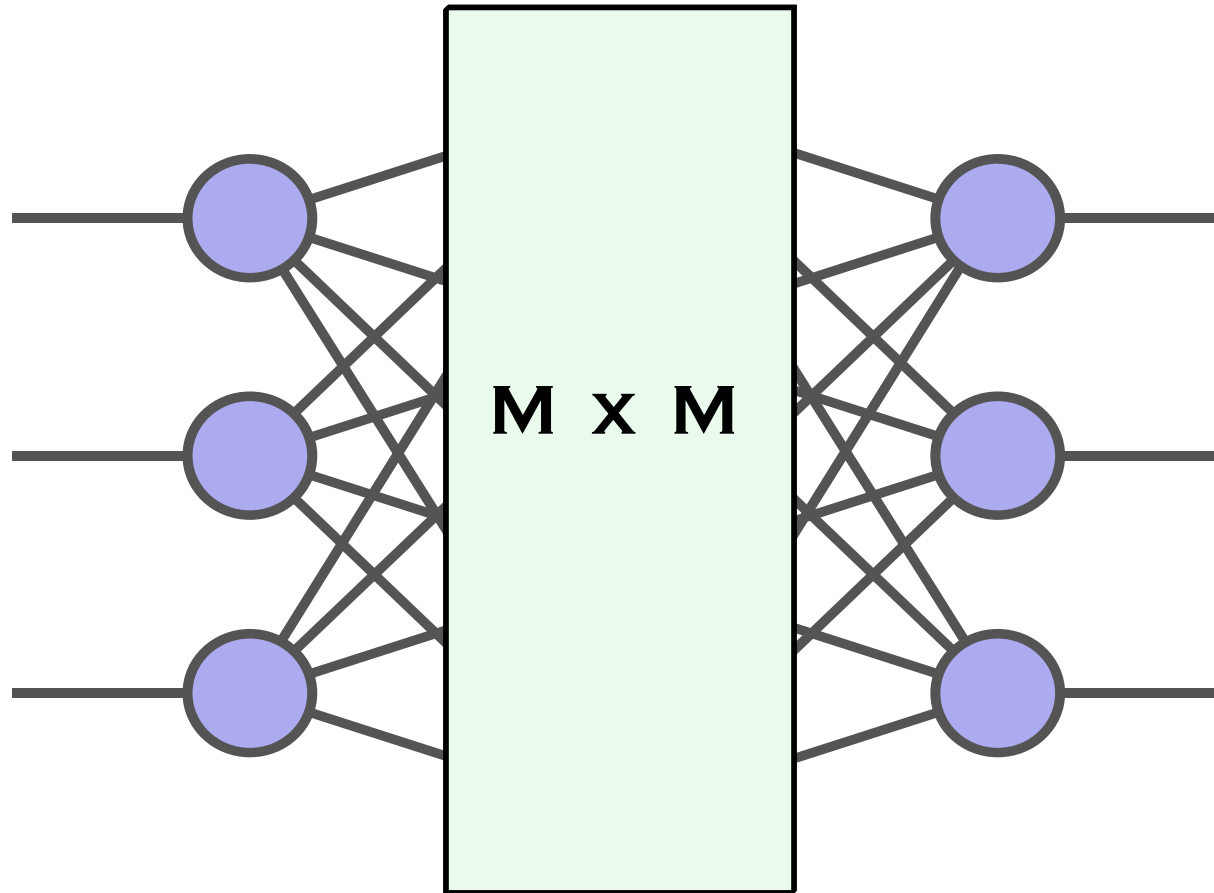


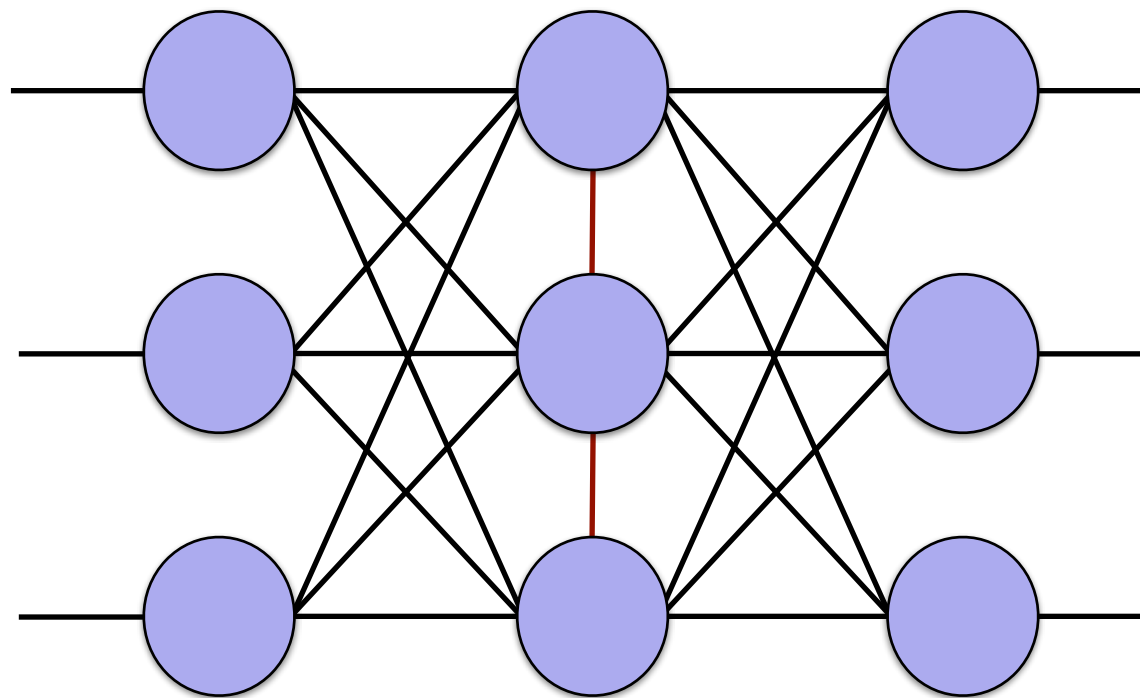
d^N



$N d B^2$

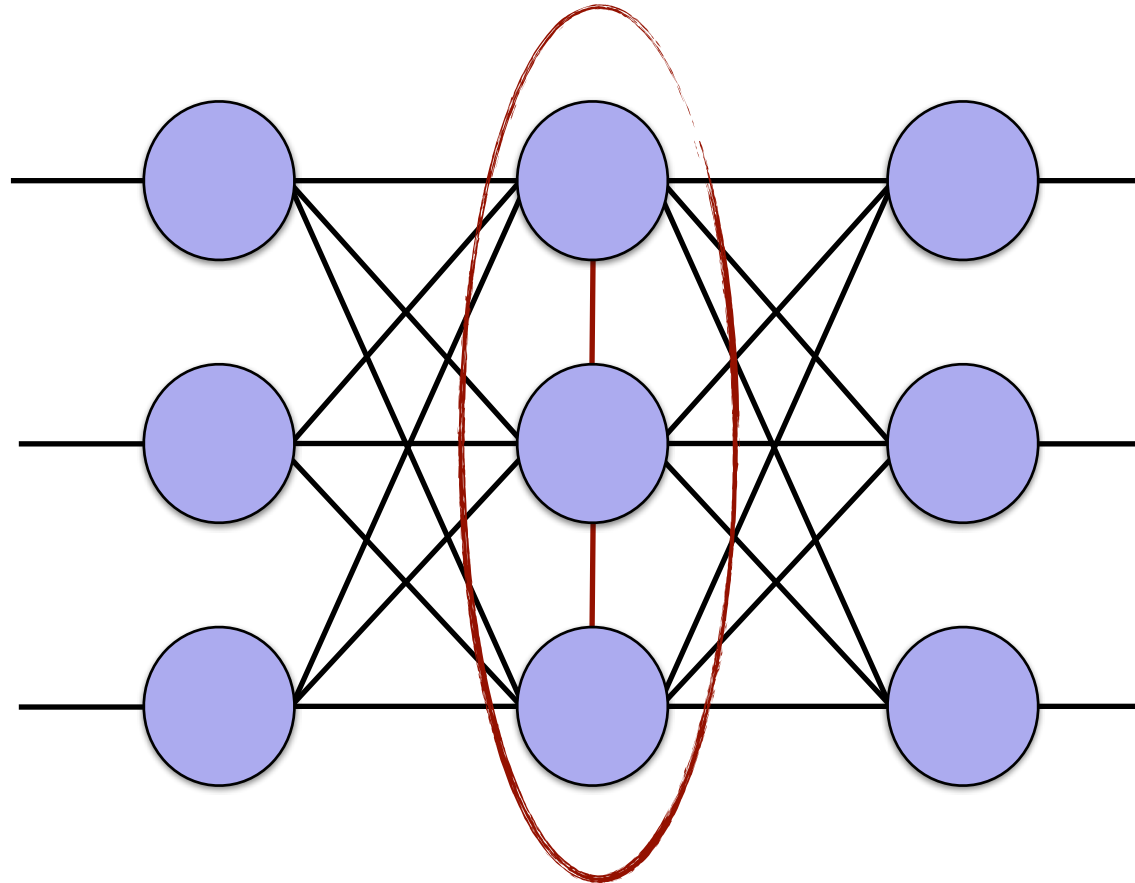




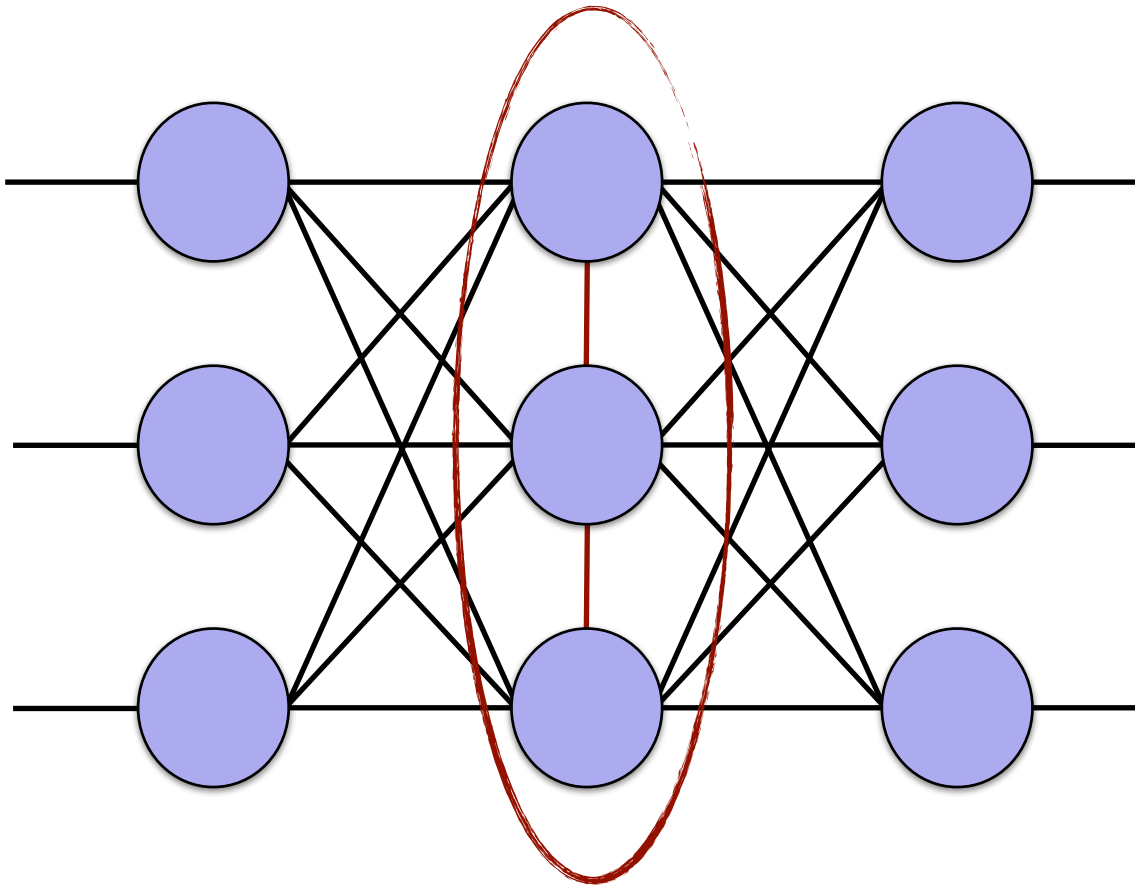


Tensorizing NNs

Compress

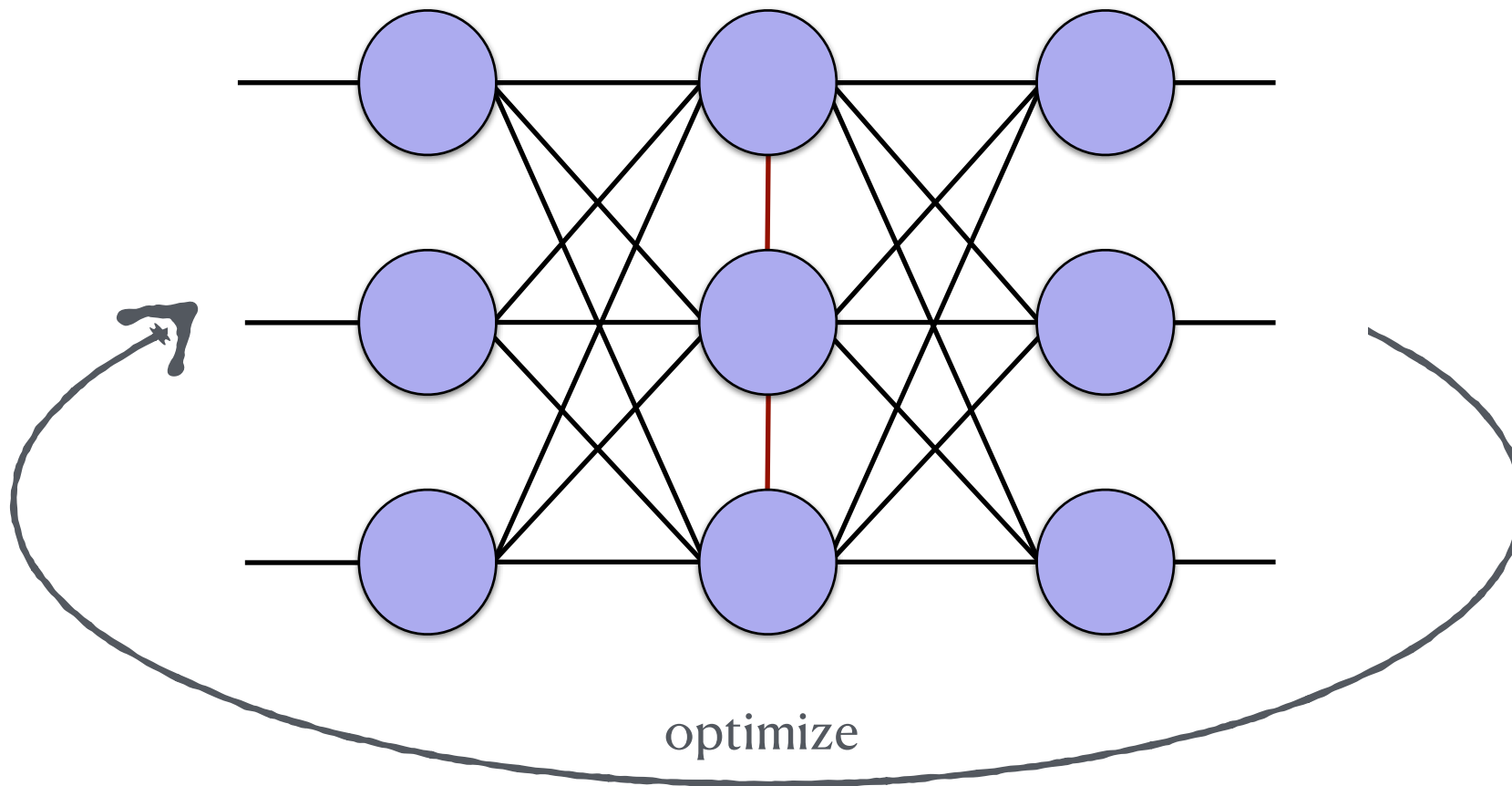


Tensorizing NNs



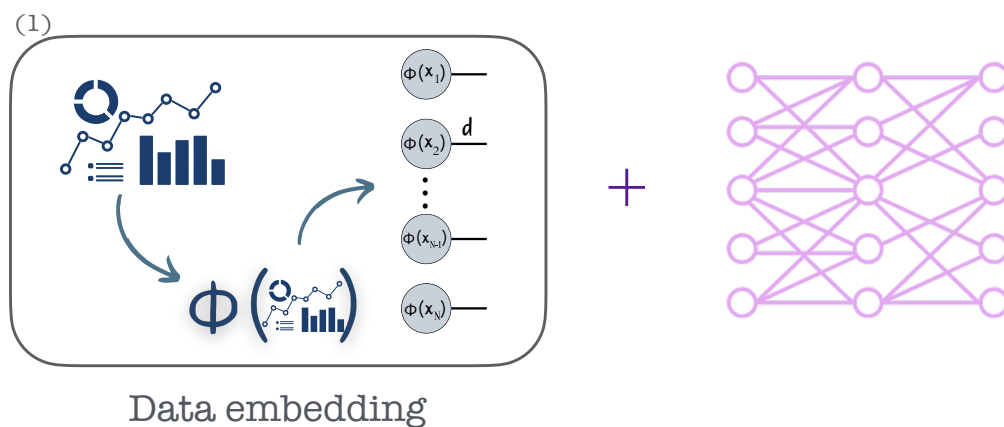
- Matrix Product State (Tensor-Train)
[1509.06569]
- Matrix Product Operator
[10.1103/PhysRevResearch.2.023300]
- Tensor Ring Decomposition
[1606.05535]
- Tucker Decomposition
[10.1109/TSUSC.2024.3425962]
- CP / PARAFAC Decomposition
[1412.6553]

Tensorizing NNs

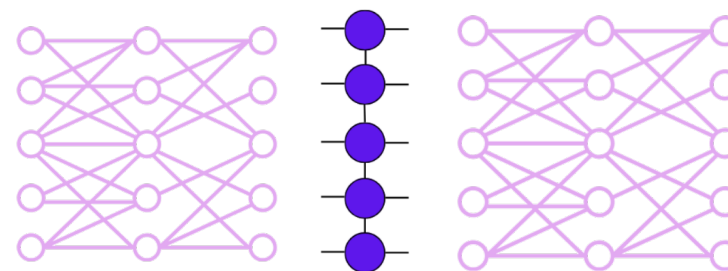


Tensor Networks as AI models

2) Data compression with TN + NN

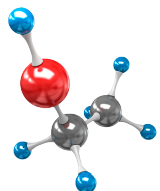
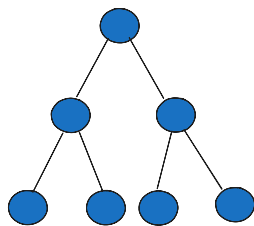


4) Hybrid TN-NN structures

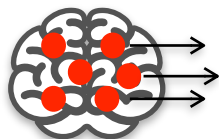


Different types of TNs

Tree TN



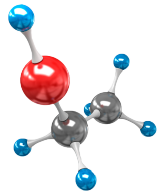
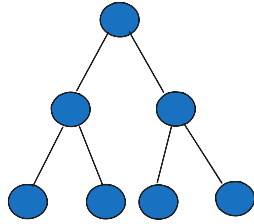
Quantum chemistry



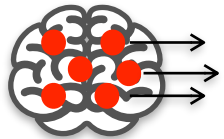
ML feature extraction

Different types of TNs

Tree TN

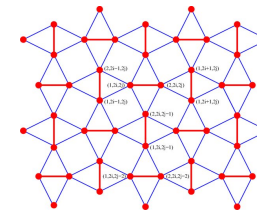
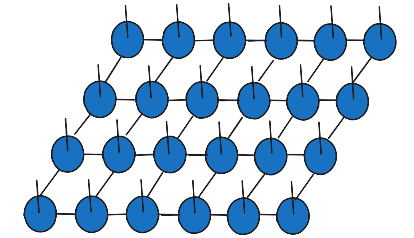


Quantum chemistry

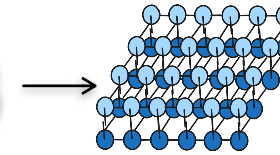


ML feature extraction

PEPS (2D) TN



quantum lattice models



classification

Materials

- An Introductory Course on Tensor Networks [[1603.03039](#)]
- A Practical Introduction to Tensor Networks: Matrix Product States and Projected Entangled Pair States [[1306.2164](#)]
- `tn4ml`: Tensor Network Training and Customization for Machine Learning [[2502.13090](#)]
- Supervised Learning with Quantum-Inspired Tensor Networks [[1605.05775](#)]
- Tensorizing Neural Networks [[1509.06569](#)]
- Tensor Networks Meet Neural Networks [[2302.09019](#)]